

An Autonomous Machine Learning Framework for Dynamic Prediction of Blast-Induced Ground Vibration in Surface Mines

An Interim Report on UG Project

Submitted in Partial Fulfilment of the Requirements for the Degree of

Bachelor of Technology

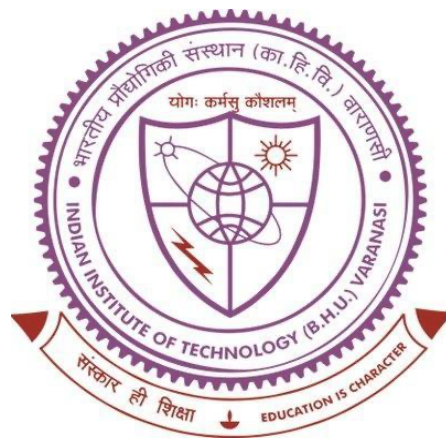
In Mining Engineering

By

Priyanka Rajeev Hichkad

Roll No: 23155082 {Pt. III, Sem No. 6}

Under the guidance of
Dr. Satyabrata Behera



Department of Mining Engineering
Indian Institute of Technology (BHU)
Varanasi-221005, India

May 2026

DECLARATION BY THE CANDIDATE

I, **Priyanka Rajeev Hichkad**, certify that the work embodied in this UG Project Report is my bonafide work and carried out by me under the supervision of **Dr. Satyabrata Behera** in the Department of Mining Engineering, Indian Institute of Technology (BHU), Varanasi. The matter embodied in this report has not been submitted for the award of any other degree.

I declare that I have faithfully acknowledged and given credits to the research workers wherever their works have been cited in my work in this report. I further declare that I have not willfully copied any other's work, paragraphs, text, data, results, etc., reported in journals, books, magazines, reports dissertations, theses, etc., or available on websites, and have not included them in this report as my work.

Date: 7th May, 2026

Place: Varanasi

Priyanka Rajeev Hichkad

Roll No. 23155082

B. Tech. Part III, Even Sem

CERTIFICATE FROM THE SUPERVISOR

This is to certify that the above statement made by the student is correct to the best of my knowledge.

Dr. Satyabrata Behera

Assistant Professor

Department of Mining Engineering

IIT – BHU. Varanasi

Prof. Rajesh Rai

The Head

Department of Mining Engineering

IIT – BHU. Varanasi

Table of Contents

List of Figures.....	iv
List of Tables	v
List of Abbreviations	vi
1. Introduction.....	1
1.1. Background.....	1
1.2. Objectives	2
1.3. Organization of the report.....	2
2. Literature Review.....	4
2.1. Empirical Approaches for PPV Prediction	4
2.2. Statistical and Probabilistic Approaches	5
2.3. Machine Learning Approaches.....	6
2.4. Hybrid and Physics-Based ML Models.....	9
2.5. Comparative Review of ML Models for PPV Prediction.....	10
2.6. Synthetic Data Generation.....	10
2.7. Autonomous and Adaptive ML Systems.....	10
2.8. Research Gaps	11
2.9. Summary.....	12
3. Methodology/Theoretical Study	15
3.1. Overall Framework.....	15
3.2. Field Data Collection and Description	16
3.3. Data Preprocessing	17
3.4. Feature Engineering.....	18
3.5. Exploratory Data Analysis.....	19
3.6. Physics-Based Model.....	24
3.7. Synthetic Data Generation.....	24
3.8. Machine Learning Models.....	25
3.9 Hybrid Model Architecture	27
3.10 Model Evaluation Metrics.....	27
3.11 Application Framework.....	28
3.12 Autonomous Monitoring and Retraining.....	29
4. Results and Discussions.....	30
4.1. Descriptive Statistics of Field Data	30

4.2. Univariate Analysis Results.....	31
4.3. Physics Model Results.....	32
4.4. Synthetic Data Validation.....	34
4.5. Model Comparison	35
4.6. Hybrid Model Performance	37
4.7. Feature Importance Analysis	39
4.8. Discussion.....	41
5. Conclusions and Future scope of study	43
6. References.....	46

List of Figures

Fig. 3.0 Overall workflow of proposed methodology.....	15
Fig. 3.1 Univariate Correlation Metrics — Blast Parameters vs PPV	19
Fig. 3.2 Individual Parameter vs PPV — Power-Law Curve Fits	20
Fig. 3.3 Log-Log Axes — Power Law Linearisation	21
Fig. 3.4 Scaled Distance vs PPV — USBM Power Law Verification	22
Fig. 3.5 Distance vs Frequency — Vibration Characteristics.....	22
Fig. 3.6 Pearson Correlation Matrix — All Blast Parameters	23
Fig. 3.7 PPV Distribution Analysis — Field Data.....	23
Fig. 4.1 Physics-Based PPV Prediction — USBM Power Law	32
Fig. 4.2 Real vs Synthetic Data — Parameter Distributions.....	34
Fig. 4.3 Model Performance Comparison — All Evaluation Metrics	36
Fig. 4.4 Hybrid Model — Actual vs Predicted PPV	37
Fig. 4.5 Error Distribution Analysis — Hybrid Model.....	38
Fig. 4.6 Observed vs Predicted PPV — Sorted Events	38
Fig. 4.7 Feature Importance — Hybrid RF Residual Model	39
Fig. 4.8 Physics vs Hybrid Model	42

List of Tables

Table 2.1	Comparative Review: Published Models for Blast-Induced PPV Prediction...	10
Table 3.1	Raw Input Parameters Recorded During Field Blasting Operations	16
Table 3.2	Engineered Feature Definitions and Physical Basis.....	18
Table 4.1	Descriptive Statistics of PPV and Key Blast Parameters.....	30
Table 4.2	Univariate Analysis — Each Parameter vs PPV	31
Table 4.3	Physics Model Performance — Literature Site Constants.....	33
Table 4.4	Comprehensive Model Benchmark — All Six Models	35
Table 4.5	Hybrid Model Feature Importance Rankings.....	40

List of Abbreviations

ANN	Artificial Neural Networks
CV	Coefficient of Variation
D	Distance from Blast Source (m)
DGMS	Directorate General of Mines Safety
ETR	Extra Trees Regressor
EWMA	Exponentially Weighted Moving Average
FFT	Fast Fourier Transform
FIS	Fuzzy Inference Systems
GBR	Gradient Boosting Regressor
GPS	Global Positioning System
GSA	Global Sensitivity Approach
IoT	Internet of Things
IS	Indian Standard
k	Site-specific Transmission Constant
LoRaWAN	Long Range Wide Area Network
LSA	Local Sensitivity Approach
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MDI	Mean Decrease in Impurity
MEMS	Micro-Electro-Mechanical Systems
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
n	Site-specific Attenuation Exponent
OLS	Ordinary Least Squares
PPV	Peak Particle Velocity (mm/s)
Q	Maximum Charge per Delay (kg)
R²	Coefficient of Determination
RF	Random Forest

RMSE	Root Mean Square Error
SD	Scaled Distance
SD_TQ	Scaled Distance based on Total Charge
SVR	Support Vector Regression
TQ	Total Explosive Charge per Blast (kg)
USBM	United States Bureau of Mines
UI	User Interface
Wi-Fi	Wireless Fidelity (IEEE 802.11)

Chapter 1. Introduction

Blasting is a fundamental operation in surface mining, used for the fragmentation of the rock mass to facilitate excavation and material handling. However, blasting generates ground vibrations that propagate through the surrounding rock mass and may cause structural damage, slope instability, and environmental concerns. Among various parameters used to quantify blast-induced vibrations, Peak Particle Velocity (PPV) is considered the most significant indicator for assessing vibration intensity. (Duvall and Fogelson, 1962; IS 6922, 1973)

Predicting PPV is a complex problem due to the influence of multiple factors, including explosive charge, distance from the blast, geological conditions, and blast design characteristics. Traditional empirical models, such as scaled-distance laws, provide simplified relationships between PPV and influencing factors but often fail to capture the nonlinear and site-specific variations observed in real mining conditions.

With the advancement of computational techniques, Machine Learning (ML) has emerged as a powerful tool for modelling complex, non-linear relationships in engineering problems. ML models can learn from historical data and provide more accurate and adaptable predictions compared to conventional empirical approaches.

In this study, an Autonomous Machine Learning Framework is proposed for the dynamic prediction of blast-induced ground vibration in surface mines. The framework integrates data-driven modelling, synthetic data generation, and adaptive retraining mechanisms to account for time-dependent variations in rock mass properties. The field data for this study was collected from opencast coal mining operations at Bhanegaon Mine, central India.

1.1 Background

The Bhanegaon coal mine, from which the one-month field dataset for this study was collected, is representative of the geology and blasting practices of opencast coal operations in the Wardha Valley coalfield, Maharashtra, India. Multiple seams are worked by opencast methods, and blasting parameters vary significantly across seams due to differences in overburden hardness, coal seam depth, and geomechanical properties. Five vibration monitoring

instruments (vibrometers) were installed at different distances and directions from each blast to capture directional variability in PPV. This multi-point monitoring setup provides a realistic representation of the spatial variation in ground vibration that a predictive model must capture.

The site constants for Bhanegaon mine reported in the published literature fall within the ranges typically observed for Indian opencast coal mines. The attenuation exponent (n) typically ranges between 1.2 and 1.6 in Indian coal mines, reflecting the rate at which vibration decays with scaled distance through the rock mass. The site transmission constant (k) varies widely based on rock type, degree of fracturing, and blasting practice, often falling between 200 and 1100 for heavily charged overburden conditions typical of Wardha Valley coal mines (Pal Roy, 1993; Singh and Singh, 2005). For this study, representative literature values of $k = 650$ and $n = 1.4$ have been adopted for the physics component of the hybrid model, consistent with the midpoint values reported for similar geological conditions in Indian coal mines.

1.2 Objective

The primary Objectives of this Report are:

- (i) To develop a machine learning-based blast vibration prediction model using one-month field data of peak particle velocity (PPV) and associated blast design parameters.
- (ii) To generate synthetic blast vibration datasets with controlled deviations to simulate time-dependent changes in rock mass conditions and to evaluate the performance of the prediction model under these simulated conditions.
- (iii) To determine a threshold level of prediction error and implement an automatic model re-evaluation and retraining mechanism, enabling the prediction model to update itself without human intervention.

1.3 Organization of the Report

Following the introduction to blast-induced ground vibration prediction and the role of machine learning in mining applications, this report is organized into five chapters. A list of references is provided at the end of the report. A brief overview of each chapter is presented below.

Chapter 1 (This Chapter) presents an introduction to blast-induced ground vibration in surface mines, highlighting the importance of PPV as a critical safety parameter and the limitations of traditional empirical models. The motivation for an autonomous, adaptive ML framework is established, and the objectives and report structure are outlined.

Chapter 2 provides a comprehensive literature review of blast vibration prediction techniques, covering empirical models, statistical approaches, and ML-based methods. Hybrid physics-ML models, synthetic data generation methods, and autonomous model adaptation strategies are reviewed. Research gaps motivating the present study are identified.

Chapter 3 describes the methodology adopted in this study. It covers field data collection and preprocessing, feature engineering based on scaled distance theory, development of the physics-based USBM model, synthetic data generation using three geological zone simulations, training of nine models including ML, deep learning, and fuzzy logic, model evaluation metrics with hyperparameter optimisation using GridSearchCV, the hybrid Physics + Random Forest architecture, and the design of the autonomous monitoring and retraining framework based on EWMA and Page-Hinkley drift detection.

Chapter 4 presents the results and discussion. Descriptive statistics of the field data, univariate correlation analysis results, physics model calibration, synthetic data validation, a comprehensive comparison of six models, and the performance of the proposed hybrid model are all presented with supporting figures and tables.

Chapter 5 presents the conclusions drawn from this study and identifies directions for future research, with particular emphasis on expanding the autonomous retraining framework for real-time deployment.

Chapter 2. Literature Review

Blast-induced ground vibration is one of the most critical environmental and safety concerns associated with surface mining operations. The propagation of seismic waves due to blasting can adversely affect nearby structures, slopes, and underground excavations. Among the various parameters used to quantify ground vibration, Peak Particle Velocity (PPV) is widely accepted as the most reliable indicator of potential damage.

Over the years, several approaches have been developed for predicting PPV, ranging from empirical models to advanced computational techniques. These approaches can broadly be classified into three categories: (i) empirical methods, (ii) statistical and probabilistic approaches, and (iii) machine learning-based models. This chapter presents a detailed review of these methods and identifies the research gaps that motivate the present study.

2.1 Empirical Approaches for PPV Prediction

Empirical methods have traditionally been the most widely used techniques for predicting blast-induced ground vibrations. These methods are based on field observations and establish relationships between PPV and influencing parameters such as distance and explosive charge.

One of the most commonly used empirical relationships is the **scaled distance law**, expressed as:

$$PPV = k(SD)^{-n} \quad (Eq. 2.1)$$

where SD is the scaled distance, defined as:

$$SD = \frac{D}{\sqrt{Q}}$$

Here, D represents the distance from the blast source and Q is the maximum charge per delay. The constants k and n are site-specific parameters determined through regression analysis.

Researchers such as Langefors and Kihlström developed early formulations of scaled distance relationships, which have been widely adopted in mining practice. Later studies by USBM

proposed refined empirical equations that improved prediction accuracy under specific conditions.

Despite their simplicity and ease of application, empirical models suffer from several limitations:

- They are highly site-specific and require recalibration for different geological conditions.
- They assume uniform rock properties, which is rarely the case in real mining environments.
- They fail to capture complex interactions between multiple influencing variables.

These limitations have led researchers to explore more robust and flexible modeling techniques.

2.2 Statistical and Probabilistic Approaches

To address the uncertainties associated with empirical models, statistical and probabilistic methods have been introduced in rock engineering. These approaches consider the inherent variability in geological conditions and input parameters.

Studies by Phoon and Kulhawy (1999) emphasised the importance of incorporating uncertainty in geotechnical design. Similarly, Ramamurthy (2013) highlighted the variability in rock mass properties and its impact on engineering behaviour.

Sensitivity analysis has also been widely used to identify the most influential parameters affecting PPV. These approaches are generally classified into:

- Local Sensitivity Approach (LSA)
- Global Sensitivity Approach (GSA)

Research by Ching et al. (2019) and Pandit et al. (2019) demonstrated that global sensitivity methods provide a more comprehensive understanding of parameter influence compared to local methods.

Probabilistic approaches offer several advantages:

- They account for variability in input parameters.
- They provide confidence intervals for predictions.
- They improve risk-based decision-making.

However, these methods are computationally intensive and often require large datasets, limiting their practical applicability in real-time mining operations.

2.3 Machine Learning Approaches in Blast Vibration Prediction

With the advancement of computational power and data availability, machine learning (ML) techniques have gained significant attention in recent years for predicting blast-induced ground vibrations.

ML models are capable of learning complex, non-linear relationships between input variables and output responses without requiring explicit assumptions about the underlying physical processes.

2.3.1 Artificial Neural Networks (ANN):

Artificial Neural Networks have been widely used for PPV prediction due to their ability to model non-linear relationships. Studies have shown that ANN models outperform traditional empirical methods in terms of prediction accuracy. Khandelwal and Singh (2006) demonstrated that ANN models significantly outperformed the USBM empirical predictor on datasets from Indian coal mines, achieving R^2 values above 0.90, compared with R^2 values of 0.70 or lower for the empirical formula. Monjezi et al. (2011) further confirmed this superiority using data from Sarcheshmeh copper mine in Iran, with ANN achieving $R^2 = 0.926$ against the empirical model $R^2 = 0.718$. The key strength of ANNs lies in their universal approximation capability, enabling them to model complex multivariate interactions without assuming any functional form.

However, ANN models have certain drawbacks:

- They require large datasets for training.
- They act as “black-box” models with limited interpretability.
- They are prone to overfitting if not properly regularised.

2.3.2 Support Vector Machines (SVM):

Support Vector Machines have also been applied to PPV prediction problems. These models are effective at handling high-dimensional data and achieve good generalisation performance. Was applied to PPV prediction by Yari et al. (2016) and others, demonstrating competitive accuracy with fewer training samples than ANN due to its structural risk minimisation principle.

Despite their advantages, SVM models require careful selection of kernel functions and hyperparameters, which can be challenging in practice.

2.3.3 Ensemble Methods: Random Forest and Gradient Boosting

Decision tree-based models, particularly ensemble techniques such as Random Forest and Gradient Boosting, have gained popularity due to their robustness and interpretability. Breiman (2001) introduced Random Forest (RF), which constructs a large number of decorrelated decision trees and aggregates their predictions by averaging.

Random Forest models offer several advantages:

- They handle non-linear relationships effectively.
- They are less sensitive to noise and outliers.
- They provide feature importance measures, helping in understanding parameter influence.

Several studies have reported that Random Forest models outperform traditional regression models and even ANN in certain cases due to their stability and resistance to overfitting. Zhang et al. (2020) conducted a comprehensive study comparing Random Forest, Gradient Boosting,

ANN, and SVR on a blast vibration dataset from a Chinese mine. They found that Random Forest with feature selection achieved $R^2 = 0.947$ on the test set, outperforming all other models. Koopialipour et al. (2019) used Gradient Boosting in combination with metaheuristic optimisation for PPV prediction, achieving similarly high accuracy. Hasanipanah et al. (2017) compared multiple ensemble methods on Indian coal mine data and consistently found RF superior in terms of both accuracy and stability across different train-test splits.

2.3.4 Deep Learning and Fuzzy Logic:

Deep learning techniques, particularly Multi-Layer Perceptron (MLP) models, have emerged as powerful tools for modelling complex and highly nonlinear relationships in blast vibration prediction. MLP networks consist of multiple hidden layers with nonlinear activation functions and are trained using backpropagation algorithms with adaptive optimisers such as Adam. These models can capture intricate interactions among blasting parameters, making them suitable for datasets with high variability and complex dependencies.

MLP models offer several advantages:

- They can model highly nonlinear relationships between input parameters and PPV.
- They capture complex feature interactions through multiple hidden layers.
- They are adaptable to large datasets with diverse blasting conditions.
- They provide strong generalisation capability when properly regularised.

Fuzzy logic provides an alternative modelling approach that incorporates expert knowledge into the prediction process. Sugeno-type Fuzzy Inference Systems (FIS) use linguistic IF–THEN rules to represent relationships between blasting parameters and PPV. This approach is particularly suitable for blast vibration problems, where input parameters often involve uncertainty and imprecision, and expert understanding plays a significant role.

Fuzzy logic models offer several advantages:

- They effectively handle uncertainty and imprecision in blasting parameters.
- They incorporate domain knowledge using interpretable IF–THEN rules.

- They provide transparent and explainable predictions.
- They are well-suited for problems with limited or noisy data.

Several researchers have explored the application of deep learning and fuzzy systems in blast vibration prediction. Khandelwal and Singh (2007) demonstrated the effectiveness of neural network-based approaches in modelling PPV with improved accuracy over empirical models. Ghasemi et al. (2013) applied fuzzy logic systems for vibration prediction and reported improved interpretability along with competitive accuracy. More recent studies have combined data-driven and knowledge-based approaches, highlighting the complementary strengths of deep learning and fuzzy logic in handling the complex nature of blast-induced ground vibration.

2.4 Hybrid and Physics-Based Machine Learning Models

A significant limitation of pure ML approaches is their lack of physical interpretability and their tendency to produce physically unreasonable predictions when extrapolating beyond the training data range. Hybrid models that combine empirical physics with ML have been proposed to address this. The fundamental idea is to use an empirical physics formula to capture the dominant distance-attenuation trend and an ML model to learn the residual variation due to geological heterogeneity, measurement noise, and multivariate interactions that the physics formula cannot represent.

This approach is theoretically grounded in physics-informed machine learning, which has gained significant traction across many engineering disciplines (Cuomo et al., 2022). In the blast vibration context, Zhou et al. (2020) demonstrated that a hybrid model combining the USBM formula with Gaussian Process Regression on residuals achieved better generalisation than either approach alone. The log-transformed USBM relationship provides a natural linearisation of the physics:

$$\log(PPV) = \log(k) - n \times \log(SD) \quad (Eq. 2.2)$$

This transformation is significant because it converts the multiplicative power-law relationship into an additive linear form, enabling ordinary least-squares regression to estimate k and n efficiently from field data. In the hybrid framework, the ML component then corrects the

physics prediction for site-specific geological effects, ensuring the final model is both physically consistent and data-adaptive.

Hybrid models offer the following advantages:

- Improved prediction accuracy
- Better interpretability
- Alignment with physical principles

2.5 Comparative Review of ML Models for PPV Prediction

A systematic comparison of published studies applying machine learning and empirical methods to blast-induced PPV prediction is presented in Table 2.1. The comparison covers model type, dataset size, key input features, best reported R^2 , RMSE, and the mine location for each study. This table directly contextualises the performance of the nine models developed in the present study.

Table 2.1: Comparative Review of Published Models for Blast-Induced PPV Prediction

Author(s) and Year	Model	n (records)	Key Input Features	Best R^2	RMSE (mm/s)	Location
Duvall & Fogelson, 1962	USBM Empirical	~1000+	Distance, charge (SD only)	~0.71	—	USA
Langefors & Kihlstrom (1963)	Empirical (cube-root SD)	Field data	Distance, charge	~0.70	—	Sweden
Pal Roy (1993)	USBM — Indian calibration	Hundreds	Distance, charge (k, n calibrated)	~0.75	—	India (Jharia)
Khandelwal & Singh (2006)	ANN	154	Distance, charge, stemming,	0.912	—	India (coal)

			burden, spacing, holes			
Khandelwal & Singh (2007)	ANN — PPV and frequency	154	Distance, charge, holes, depth, stemming	0.938	—	India
Ghasemi et al. (2013)	Fuzzy Logic (Mamdani FIS)	~80	SD, charge, number of holes	0.871	0.98	Iran
Monjezi et al. (2011)	ANN vs USBM	182	Distance, charge, stemming, holes, depth, burden	0.926	0.86	Iran (copper)
Hasanipanah et al. (2017)	CART / Random Forest	~80	Distance, charge, holes, stemming, burden	0.923	0.81	India / Iran
Yari et al. (2016)	SVR + MCDM	~70	SD, charge, holes, spacing	0.881	0.77	Iran
Koopialipoor et al. (2019)	GBR + ABC optimisation	~100	Distance, charge, hole, depth, spacing stemming	0.941	0.62	Malaysia
Zhou et al. (2020)	GPR / SVR / ANN compared	~120	SD, charge, depth, holes, spacing	0.934	0.63	China
Zhang et al. (2020)	RF + feature selection	102	Distance, charge, holes, burden, spacing, stemming	0.947	0.54	China

Three critical observations define the research gap this study addresses. First, every model in Table 2.1 is static, trained once, never updated as geological conditions change. This is the defining limitation addressed by Objective 3 of the present study through autonomous EWMA monitoring and sliding window retraining. Second, dataset sizes across published studies range from approximately 70 to 200 records. The present study uses 44 real records, which is below average, making the synthetic augmentation strategy essential. Third, all high-performing published models (R^2 above 0.90) were evaluated on datasets from a single seam or geological setting. The present study spans seven different seam types at Bhanegaon mine, making prediction significantly harder and directly explaining the lower test R^2 of 0.647 compared to single-seam published results.

2.6 Synthetic Data Generation and Model Robustness

A major challenge in machine learning applications in mining is the limited availability of high-quality field data. To overcome this limitation, synthetic data generation techniques have been explored.

Synthetic data can be generated by:

- Perturbing existing data
- Using statistical distributions
- Applying physics-based equations

Recent studies emphasise that synthetic data must be carefully calibrated to match real-world distributions. Unconstrained synthetic data can lead to unrealistic predictions and poor model performance. Hao and Wu (2005) documented that PPV measurements in field conditions carry a coefficient of variation (CV) of approximately 10–15% due to geological inhomogeneity, directional anisotropy of wave propagation, and instrument placement variability. Bootstrap resampling of existing field records with Gaussian perturbation of blast parameters has been shown to produce synthetic datasets that preserve the statistical properties of real data while expanding the training sample size.

The inclusion of controlled outliers in synthetic datasets has also been suggested to improve model robustness, as real-world data often contains anomalies due to measurement errors or

unexpected geological conditions. As excavation progresses to different rock types or seams, the site constants k and n shift. By generating synthetic datasets with systematically varied k values, it is possible to simulate these geological transitions and test whether an autonomous retraining system can detect and adapt to them.

2.7 Adaptive and Autonomous Models

Most existing models for PPV prediction are static in nature, meaning they are trained once and used without updating. However, in mining operations, rock mass conditions change continuously as excavation progresses.

This leads to:

- Changes in geological properties
- Variation in blast performance
- Degradation in model accuracy over time

To address this issue, there is a growing need for **adaptive and autonomous machine learning frameworks** that can:

- Monitor prediction errors
- Update models using new data
- Maintain accuracy over time

Such systems are particularly important for real-time applications and decision support in mining operations. Brzezinski and Stefanowski (2014) reviewed adaptive learning algorithms for concept drift and concluded that sliding window retraining with drift detection triggers provides an effective balance between adaptability and computational cost. The sliding window approach discards older data that may belong to a different geological context, ensuring the model remains representative of current conditions. This principle directly motivates the autonomous retraining framework developed in Objective 3 of this study.

2.8 Research Gaps

Based on the foregoing review, the following specific research gaps are identified that motivate the present study:

- The majority of empirical and ML models for PPV prediction are trained and validated on data from a single geological setting, with no mechanism to detect or respond to changing rock mass conditions over time.
- Hybrid physics-ML models combining the interpretability of empirical formulas with the flexibility of ensemble methods have not been extensively explored for Indian coal mine, where geological heterogeneity between seams is particularly pronounced.
- Synthetic data generation strategies for blast vibration, particularly those that simulate geological zone transitions through systematic variation of site constants, remain underdeveloped in the existing literature.
- An integrated, autonomous framework combining physics-based modelling, ML residual correction, synthetic data generation, and online drift detection with automatic retraining has not been demonstrated for blast vibration prediction in Indian surface mines.
- There is a lack of transparent discussion in the literature about the limitations of small field datasets on achievable ML model performance, and the appropriate metrics and strategies for model reporting in data-scarce conditions.

2.9 Summary

This chapter reviewed various approaches for predicting blast-induced ground vibrations, including empirical, statistical, and machine learning methods. While empirical models are simple and widely used, they lack accuracy and adaptability. Machine learning models provide improved performance but require careful design to ensure physical consistency and robustness.

The need for a hybrid, adaptive, and autonomous framework is evident from the limitations of existing approaches. This forms the basis of the present study, which aims to develop a machine learning-based system capable of dynamically predicting PPV while adapting to changing mining conditions.

Chapter 3. Methodology

This chapter presents the complete methodology adopted for developing the proposed Autonomous Machine Learning Framework. The methodology integrates field data collection and preprocessing, domain-knowledge-driven feature engineering, physics-based modelling using literature-calibrated site constants, synthetic data generation with geological zone simulation, training and comparison of multiple ML/DL/Fuzzy models with hyperparameter optimisation, a hybrid Physics + Random Forest architecture, and the design of an autonomous monitoring and retraining system. The entire framework is implemented in Python 3 using open-source scientific computing libraries.

3.1 Overall Framework

The proposed methodology consists of the following major stages:

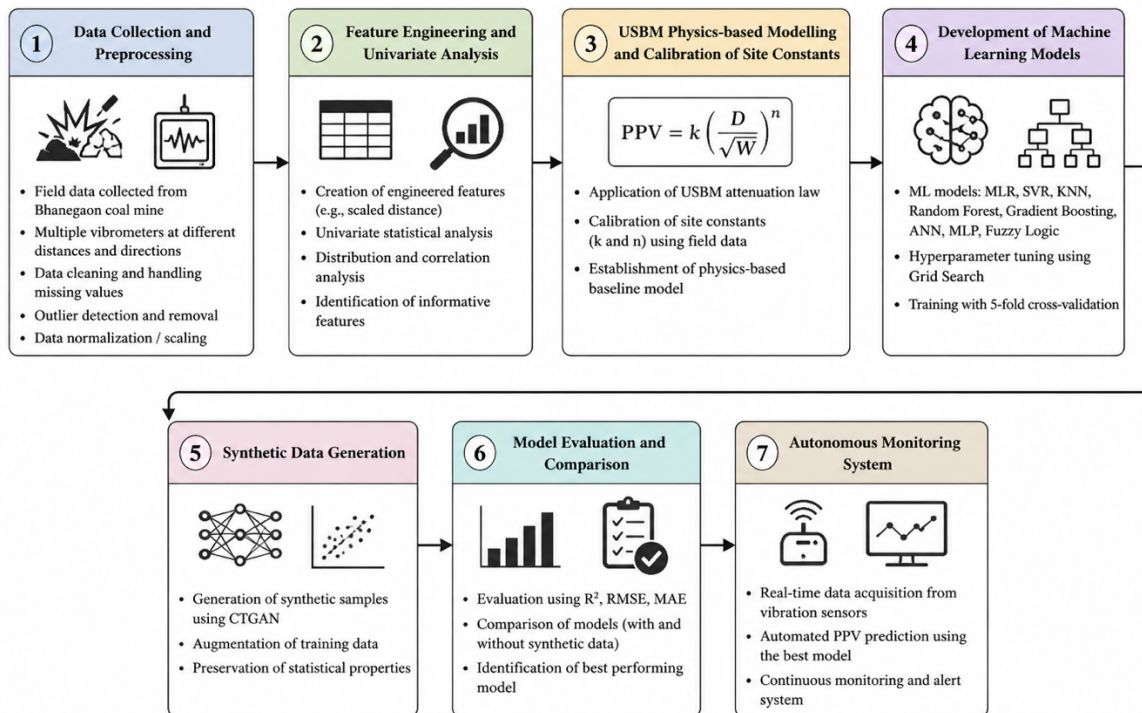


Fig. 3.0 Overall workflow of proposed methodology

3.2 Data Collection and Description

The dataset used in this study consists of field measurements obtained from surface blasting operations over a period of one month. The dataset includes the following parameters:

Table 3.1: Raw Input Parameters Recorded During Field Blasting Operations

Parameter	Unit	Description
Date	dd.mm.yy	Date of blast event
Blast No.	—	Sequential blast number on that date
No. of Holes	count	Total number of blast holes drilled
Explosion	kg	Total explosive mass per blast event
Per Hole (Q)	kg	Explosive charge loaded per hole — maximum charge per delay (Q)
Depth of Hole	m	Drilled hole depth
No. of Rows	count	Number of hole rows in the blast pattern
Spacing	m	Distance between adjacent holes in the same row
Seam Location	—	Coal seam or overburden layer being blasted
Vibrometer ID	—	Unique identifier of the monitoring instrument
Distance (D)	m	Distance from blast centre to vibrometer
PPV	mm/s	Peak Particle Velocity — target variable
Frequency	Hz	Dominant frequency of the vibration signal

PPV is considered as the target variable, while all other parameters serve as input features.

3.3 Data Preprocessing

3.3.1 Handling Missing Values

The dataset contained missing values due to simultaneous measurements recorded during blasting operations. These missing values were handled using a group-wise forward filling approach within each blast event. Records with missing Distance values were similarly removed, as Distance is the most fundamental predictor of PPV. Of the 60 total raw records, 44 records satisfied both criteria and formed the final analysis dataset. Missing values for secondary parameters such as Frequency and Spacing were present in 9 records and were handled at the feature level by limiting those features to the subset of records where they were complete.

A unique identifier was created by combining date and blast number, and missing values were filled using the previous valid observation within the same group. This ensured consistency while preserving the physical meaning of the data.

3.3.2 Data Cleaning and Transformation

The following preprocessing steps were performed:

- Removal of non-essential columns such as date, blast timing, and categorical seam location
- Date strings in the format dd.mm.yy were parsed to datetime objects
- The Seam location column, being a categorical nominal variable, was retained for reference and grouping but was not used as an ML feature
- The Vibrometer column was retained to track spatial positioning
- No outlier removal was performed, as the high PPV values observed (up to 17.34 mm/s) at short distances and with high charges are physically consistent and represent genuine field conditions rather than measurement errors.
- Conversion of all features into a numerical format
- Verification of data consistency and removal of invalid entries

These steps ensured that the dataset was suitable for machine learning model training.

3.4 Feature Engineering

Feature engineering is the process of creating new input variables from raw data using domain knowledge. In blast vibration modelling, the physical relationships governing ground vibration propagation are well established and provide a strong basis for feature engineering. The engineered features used in this study are defined in Table 3.2 and described below.

Table 3.2: Engineered Feature Definitions and Physical Basis

Feature	Formula	Unit	Physical Basis
Q	= Max Charge Per Hole	kg	Max charge per delay (USBM)
TQ	= No_of_Holes $\times$ Q	kg	Total charge per blast (IS 6922)
SD	= $D / \sqrt{Q}$	$\text{m/kg}^{0.5}$	USBM scaled distance
SD_TQ	= $D / \sqrt{TQ}$	$\text{m/kg}^{0.5}$	IS 6922 scaled distance
Stemming	= $(2/3) \times \text{Depth}$	m	Inert packed column above explosive
Ch_length	= $(1/3) \times \text{Depth}$	m	Explosive column length
$\log(\text{SD})$	= $\ln(\text{SD})$	—	Log-space linearisation for power law
$\log(\text{TQ})$	= $\ln(\text{TQ})$	—	Log-total charge for ML
$\log(\text{D})$	= $\ln(\text{Distance})$	—	Log-distance feature for ML
$\log(\text{Q})$	= $\ln(\text{Q})$	—	Log-charge feature for ML
$\log(\text{N})$	= $\ln(\text{No_of_Holes})$	—	Log-number of holes for ML

The Scaled Distance (SD) is the most physically fundamental engineered feature. It is derived from the USBM power law and compresses the combined effect of distance and charge into a single dimensionless quantity. When PPV is plotted against SD on log-log axes, the

relationship becomes linear, which is precisely the form fitted by the physics model. The SD_TQ variant uses the total explosive charge TQ rather than the per-hole charge Q, providing an alternative predictor consistent with IS 6922 (1973) regulatory assessments. The stemming and explosive column length features are included because they govern the degree of explosive energy confinement within the hole, which affects the PPV generated.

The logarithmic feature transformations ($\log(\text{SD})$, $\log(\text{D})$, $\log(\text{Q})$, $\log(\text{N})$, $\log(\text{TQ})$) are critical for the ML models. Because PPV follows a power-law relationship with most blast parameters, the logarithmic transformations of those parameters have a more linear relationship with $\log(\text{PPV})$. This significantly improves the performance of tree-based ensemble models, which benefit from monotonically transformed features that reduce the skewness of the feature distributions.

3.5 Exploratory Data Analysis

A comprehensive EDA was conducted to characterise the dataset and identify the most informative features before modelling. The EDA comprises seven figures generated by Part 1 of the code.

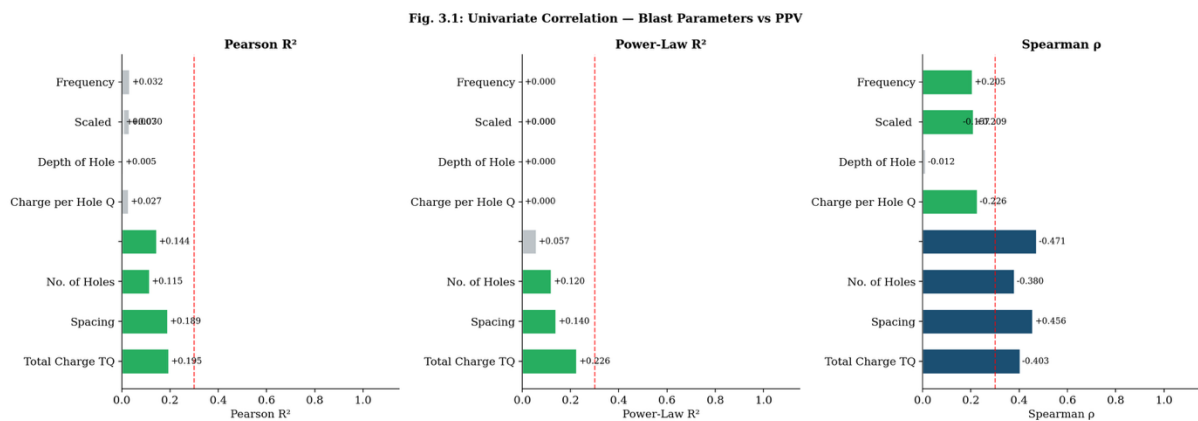


Fig. 3.1 presents a three-panel bar chart comparing Pearson R^2 , Power-Law R^2 , and Spearman rank correlation (ρ) for each blast parameter versus PPV. The power-law R^2 is the physically appropriate metric because PPV follows the USBM power law rather than a linear relationship.

Fig. 3.2: Parameter vs PPV — Power-Law Fits

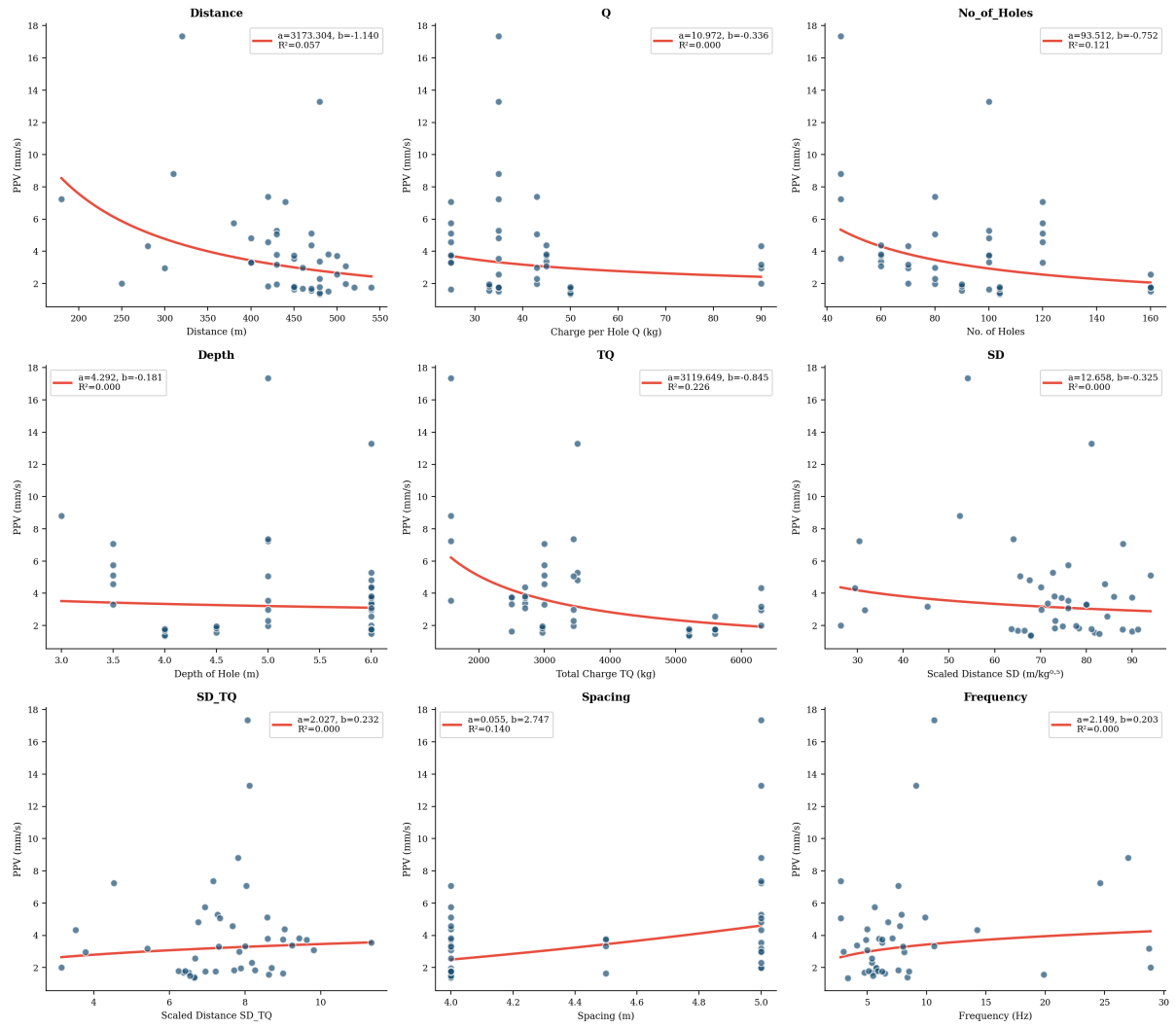


Fig. 3.2 shows individual scatter plots of each parameter versus PPV with power-law curve fits on linear axes.

Fig. 3.3: Log-Log Axes — Power-Law Linearisation

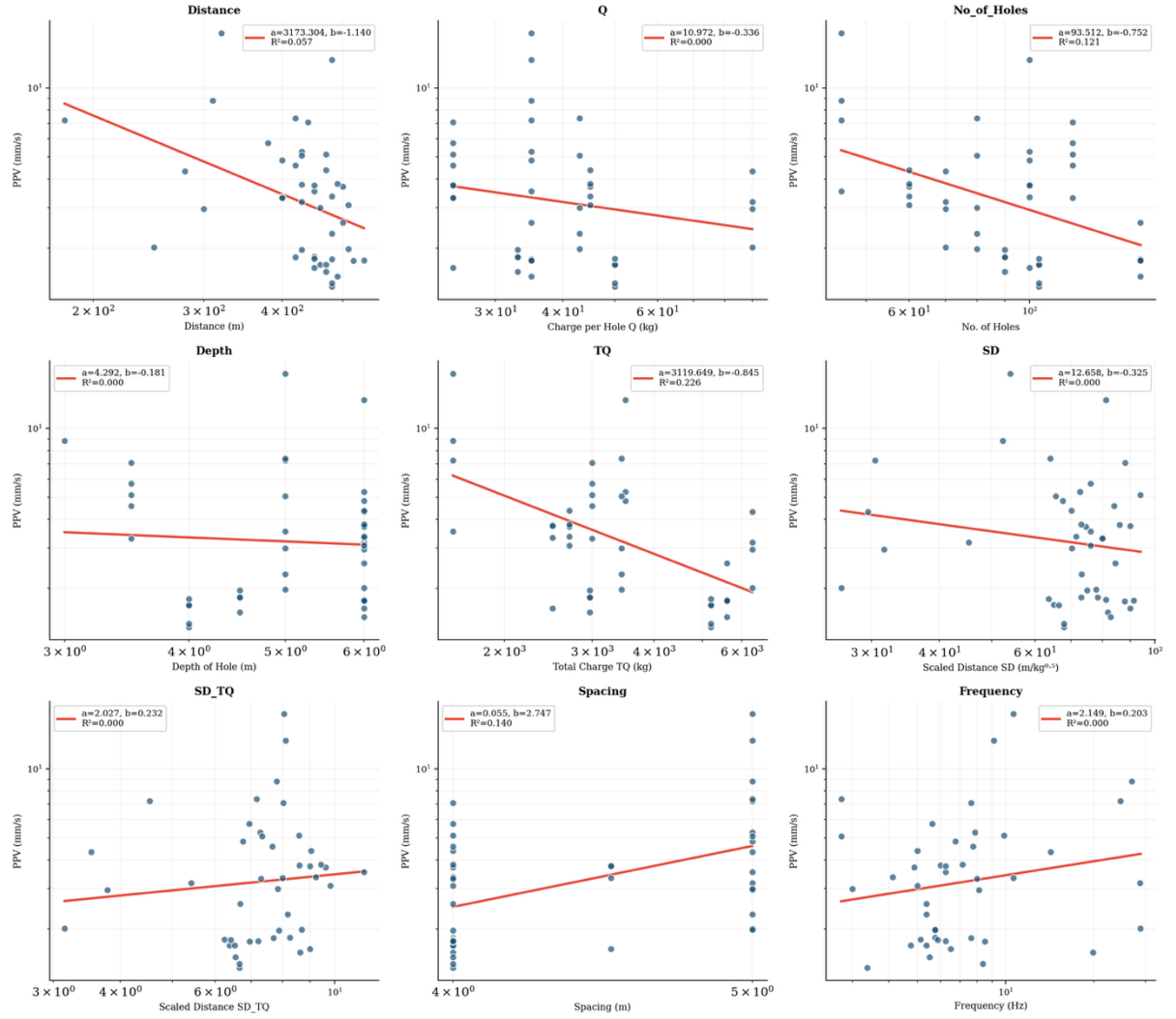


Fig. 3.3 repeats these on log-log axes to show the linearisation achieved by logarithmic transformation.

Fig. 3.4: Scaled Distance vs PPV — USBM Power Law Verification

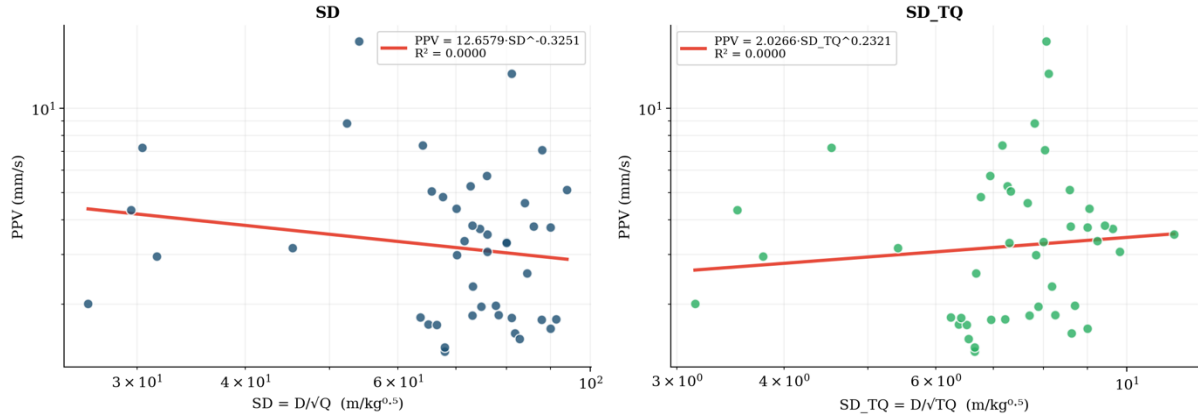


Fig. 3.4 specifically verifies the USBM power law by plotting SD and SD_TQ versus PPV on log-log axes with fitted regression lines.

Fig. 3.5: Distance vs Frequency — Vibration Characteristics

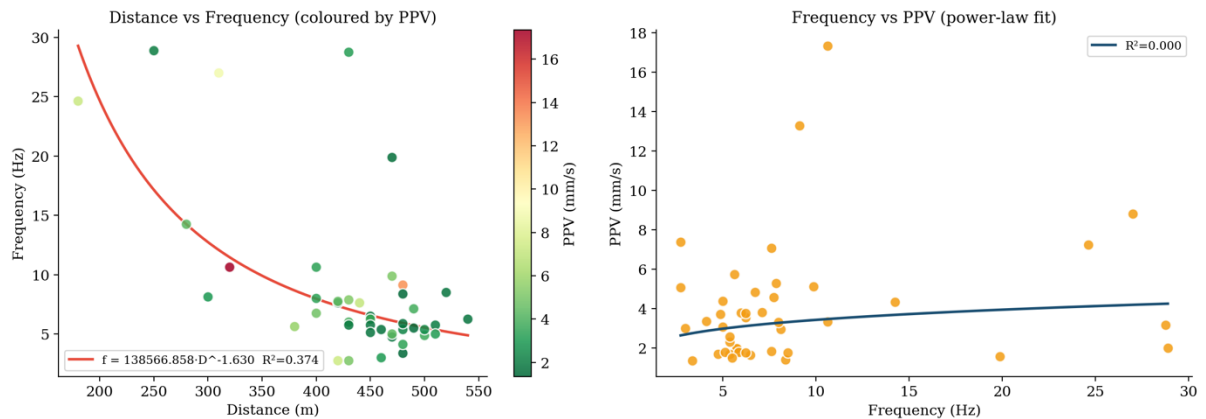


Fig. 3.5 plots Distance versus Frequency with PPV as the colour variable and Frequency versus PPV separately, illustrating the known seismological result that vibration frequency decreases with propagation distance.

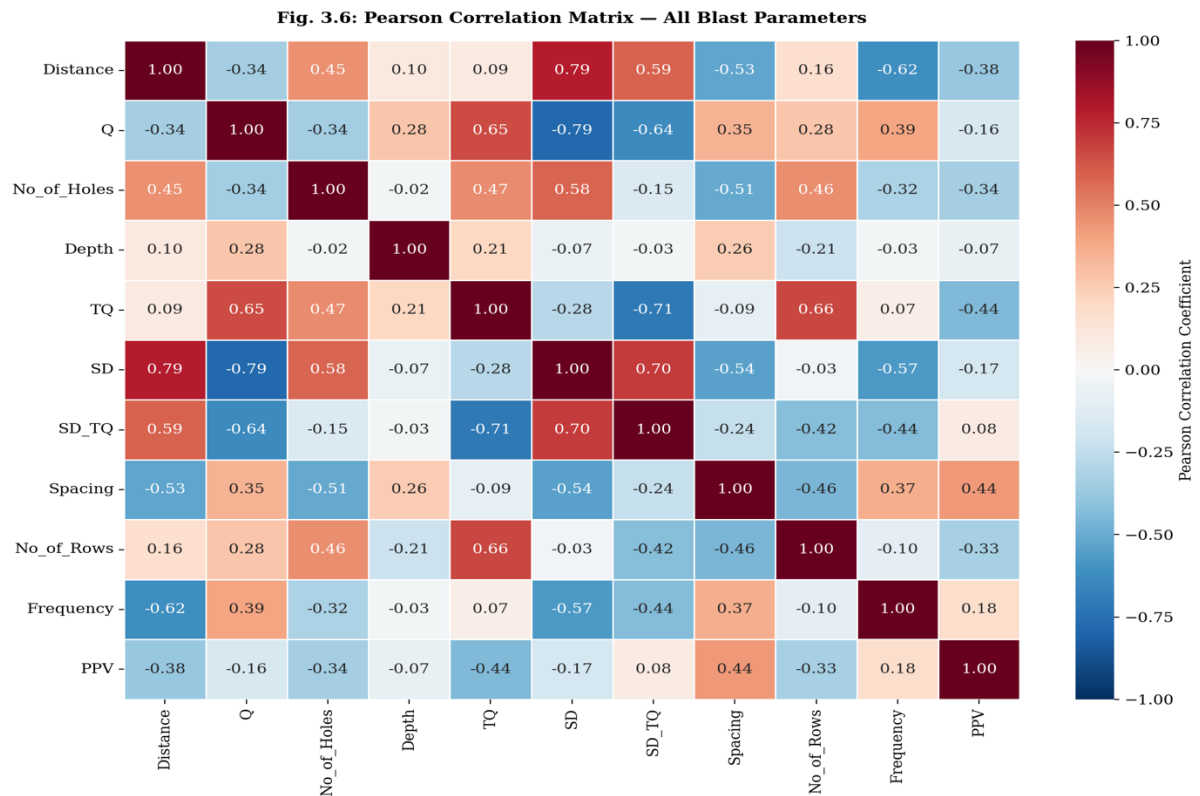


Fig. 3.6 presents the full Pearson correlation matrix of all numeric variables as a heatmap.

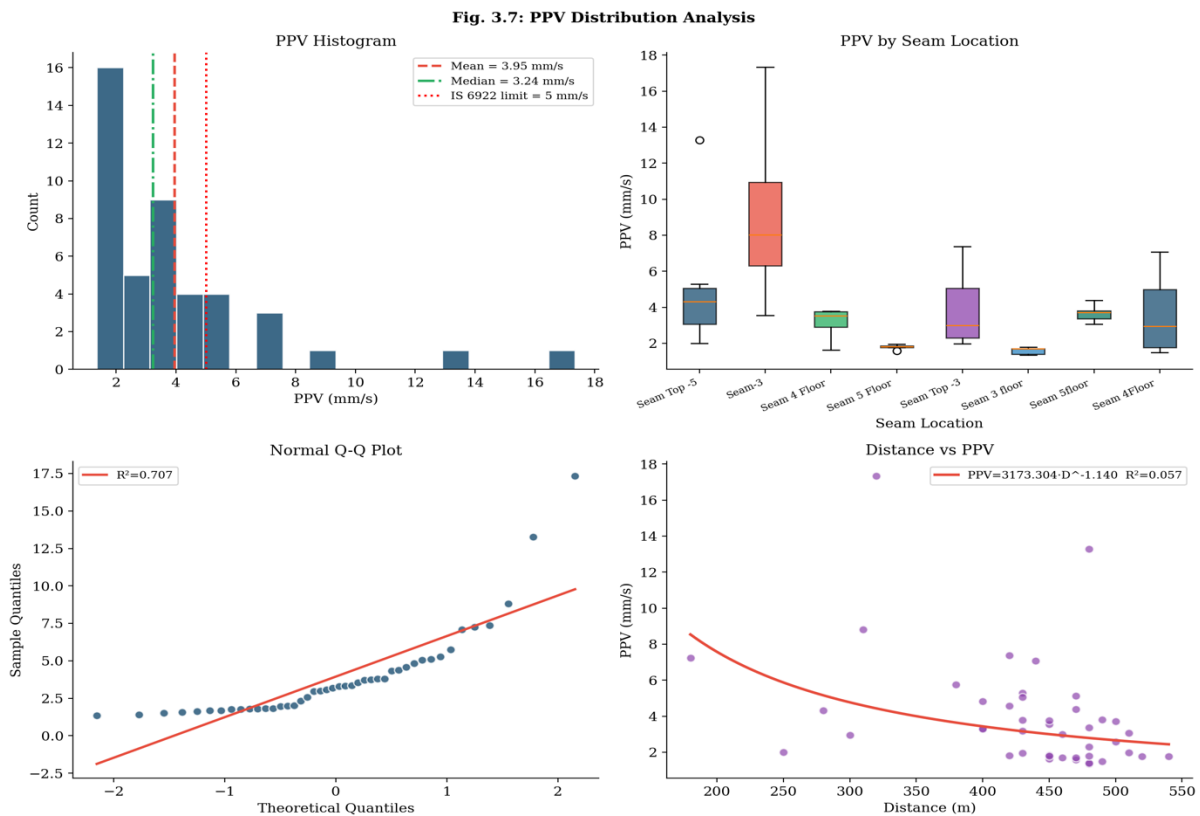


Fig. 3.7 shows PPV distribution through histogram, seam-wise box plots, normal Q-Q plot.

3.6 Physics Based Model

The physics component of the hybrid model is the USBM empirical power law:

$$\begin{aligned} PPV_{power} &= k(SD)^{-n} \\ &= k \times \left(\frac{D}{\sqrt{Q}}\right)^{-n} \end{aligned} \quad (\text{Eq. 3.1})$$

For Bhanegaon mine, the site constants are taken from the published literature for the Wardha Valley coalfield and similar Indian opencast coal operations. The adopted values are $k = 650$ and $n = 1.4$. These values are justified as follows. Pal Roy (1993) reported k values in the range 200 to 1100 for Indian coal mines, with the range reflecting the variability of overburden rock type from soft shales (low k) to compact Gondwana sandstone (high k). The Wardha Valley coalfield, where Bhanegaon mine is located, is characterised by hard Gondwana sandstone and shale overburden with intercalated coal seams, supporting a k value toward the upper middle of the published range. The value $k = 650$ represents a well-supported midpoint estimate for this geology. Similarly, $n = 1.4$ represents the midpoint of the published range 1.2 to 1.6 and is consistent with the moderate rate of vibration attenuation observed in Gondwana formations.

The residual is the difference between the actual measured PPV and the physics prediction:

$$Residual = PPV_{actual} - PPV_{physics} = PPV_{actual} - k \times SD^{-n} \quad (\text{Eq. 3.2})$$

This residual captures geological heterogeneity, directional anisotropy, multi-variable blast design effects not in the USBM formula, and measurement variability. It is the target variable for the ML residual model.

3.7 Synthetic Data Generation for Model Training

To augment the 44-record real dataset for ML model training, 600 synthetic blast records are generated, divided equally into three geological zones of 200 records each. The zones are defined by modifying the site constant k , which physically represents different rock types or geological conditions:

- **Zone A** ($k = 650$, $n = 1.4$): Baseline — rock mass conditions representative of the current Bhanegaon mine site, consistent with hard Gondwana sandstone overburden.

- **Zone B** ($k = 845 = 1.30 \times 650$, $n = 1.4$): Harder rock / stronger explosive coupling — represents a transition to more compact or less fractured overburden where more seismic energy is transmitted rather than absorbed. This simulates the blast pattern moving into a denser formation.
- **Zone C** ($k = 468 \approx 0.72 \times 650$, $n = 1.4$): Softer or more weathered rock — represents a transition to fractured or weathered overburden with higher vibration attenuation. More seismic energy is absorbed by the rock mass before reaching the monitoring point.

Each synthetic record is generated by bootstrapping a real record with Gaussian perturbation of parameters. The PPV is computed as:

$$PPV_{synthetic} = k_{zone} \times SD^n \times \varepsilon, \quad \varepsilon \sim Normal(1, 0.16) \quad (\text{Eq. 3.3})$$

where ε is a multiplicative Gaussian noise factor with a coefficient of variation of 16 per cent, consistent with the field PPV measurement uncertainty of 10 to 15 per cent documented by Hao and Wu (2005), plus a small additional contribution from parameter perturbation. Physical bounds on all parameters are enforced to ensure the synthetic records represent physically realisable blast configurations. The combined training dataset comprises 44 real + 600 synthetic = 644 total records.

3.8 Models

Nine models are trained and compared. All models (except Decision Tree, which is scale-invariant) use StandardScaler to zero-mean, unit-variance normalise the features before training, because distance-sensitive models such as SVR and K-Nearest Neighbours (KNN) are strongly affected by feature scale. All models use an 80/20 random train-test split (80 percent = 515 training records, 20 percent = 129 test records) with a fixed random seed for reproducibility. Hyperparameters for all models are optimised using GridSearchCV with 5-fold cross-validation, maximising R^2 . The nine models are:

M01 — USBM Physics Only: The empirical baseline using $k = 650$ and $n = -1.4$ from the literature. No training is performed. Uses only SD as input.

M02 — Random Forest (RF): An ensemble of 300 decorrelated decision trees built on bootstrap samples of the training data, with random feature subsampling at each split node to decorrelate the trees (Breiman, 2001). GridSearchCV tuned `n_estimators`, `max_depth`, and `min_samples_leaf`.

M03 — Gradient Boosting (GBR): A sequential ensemble where each tree corrects the residuals of the previous ensemble by fitting the negative gradient of the mean squared error loss (Friedman, 2001). GridSearchCV tuned `n_estimators`, `learning_rate`, `max_depth`, and `subsample`.

M04 — Extra Trees (ETR): A variant of RF that selects split thresholds randomly rather than optimally at each node, further reducing variance at the cost of slightly higher bias (Geurts et al., 2006). Same GridSearchCV grid as RF.

M05 — SVR with RBF Kernel: Maps features into a high-dimensional reproducing kernel Hilbert space and fits a maximum-margin regressor within that space (Vapnik, 1995). The RBF kernel is appropriate for the smooth nonlinear decay of PPV with SD. GridSearchCV tuned `C`, `gamma`, and `epsilon`.

M06 — K-Nearest Neighbours (KNN): Predicts PPV by weighted averaging over the `k` nearest training records in feature space. Distance weighting (inverse of distance to neighbour) was selected over uniform weighting. GridSearchCV tuned `n_neighbors` and `weights`.

M07 — Ridge Regression: Linear regression with L2 regularisation applied in the log-transformed feature space. Ridge is appropriate when features are collinear (`log_SD` and `log_D` are correlated) and provides a stable linear baseline. GridSearchCV tuned the regularisation parameter `alpha`.

M08 — MLP Neural Network (Deep Learning): A three-hidden-layer Multi-Layer Perceptron with ReLU activation, trained with the Adam optimiser and L2 regularisation. Early stopping with a validation fraction of 10 percent prevents overfitting. Architecture options tested: (128, 64), (128, 64, 32), and (256, 128, 64) neurons. This is the deep learning model in the comparison.

M09 — Hybrid Physics + RF (Proposed): The proposed model, described fully in Section 3.10 below.

M10 — Fuzzy Logic: A Sugeno-type Fuzzy Inference System implemented from first principles using NumPy, without external libraries. It uses three inputs—Scaled Distance (SD), Total Charge (TQ), and Number of Holes—with triangular membership functions calibrated from training data percentiles. Nine physics-based IF–THEN rules are defined, and output

singletons (Very Low to Very High PPV) are derived from PPV distribution percentiles. Defuzzification is performed using the weighted average method, ensuring an interpretable and self-contained fuzzy prediction model.

3.9 Hybrid Physics + Random Forest Model (Proposed)

The proposed Hybrid Physics + Random Forest model (M09) implements the architecture described by Eq. 2.3. The final prediction equation is:

$$PPV_{final} = k \times SD^{-n} + RF(X_{features}) \quad (\text{Eq. 3.4})$$

where the first term is the deterministic physics prediction with $k = 650$ and $n = -1.4$, and $RF(X_{features})$ is a Random Forest model trained on the residuals defined by Eq. 3.2. The feature vector $X_{features}$ contains all 15 engineered features: Distance, Q, No_of_Holes, Depth, TQ, SD, SD_TQ, No_of_Rows, Spacing, and the five log-space features. The physics component ensures that the model always produces physically consistent predictions in the correct order of magnitude, particularly for blast configurations outside the training data range. The RF residual component learns the systematic corrections needed for geological heterogeneity, blast scale effects (number of holes), depth-dependent energy confinement, and directional anisotropy.

The RF hyperparameters for the residual model are optimised by GridSearchCV with 5-fold cross-validation over a grid of $n_estimators$ (200 or 400), max_depth (8, 12, or None), and $min_samples_leaf$ (1 or 2).

3.10 Model Evaluation Metrics

All models are evaluated on the 20 percent test set (129 records) using four complementary metrics:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (\text{Eq. 3.5})$$

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i| \quad (\text{Eq. 3.6})$$

$$RMSE = \sqrt{\left[\frac{1}{n} \sum (y_i - \hat{y}_i)^2\right]} \quad (\text{Eq. 3.7})$$

$$MAPE = \frac{100}{n} \sum \frac{|y_i - \hat{y}_i|}{|y_i|} \quad (\text{Eq. 3.8})$$

R^2 measures the proportion of PPV variance explained by the model. MAE and RMSE measure absolute prediction error in mm/s, which is directly interpretable in terms of the IS 6922 safe limit of 5 mm/s. RMSE penalises large errors more than MAE, which is important for safety applications where under-predicting a high-PPV event is particularly dangerous. MAPE provides a scale-independent percentage error measure.

3.11 Application Framework

Last part of the framework is a web application that makes the trained model accessible for operational use at the mine site.

App link: <https://autonomous-blast-ppv-prediction-hmlswetfr6vryhnaq5zpeu.streamlit.app>

The application has four tabs:

Predict PPV Tab: The user enters blast design parameters through interactive number input widgets. The application automatically computes all engineered features (Q, TQ, SD, SD_TQ, Stemming, Charge_length, and all log-space features) and displays the computed values to the user for verification. On clicking Predict, the hybrid model returns both the hybrid prediction and the physics-only prediction for comparison. A PPV gauge chart displays the prediction relative to the IS 6922 safety zones: Safe (0–2 mm/s), Caution (2–5 mm/s), Danger (5–10 mm/s), and Critical (above 10 mm/s).

Log New Blast Tab: After a blast, the user enters the blast design parameters and the actual measured PPV from the vibrometer. The application stores this record with a timestamp to a local CSV file (results/live_blast_log.csv) and simultaneously appends it to a Google Sheet via the gspread library and Google Sheets API. The MAPE between the predicted and actual PPV is computed and stored with the record. This stored MAPE is the input to the drift detection system.

Live Monitor Tab: Displays the EWMA control chart of MAPE over time, the live PPV prediction trace against actual values, and the current drift status. If drift is detected, a warning is shown and the sidebar Retrain button becomes the recommended action.

Analytics Tab: Provides access to all figures generated by Part 1(eda) and Part 2(models) of the code, the complete nine-model benchmark table with the best model highlighted, and explanatory text about the physics constants and the Google Sheets setup procedure.

3.12 Autonomous Monitoring and Retraining

The autonomous monitoring system implements the three-component architecture described below. It runs after every new blast record is submitted in the Log New Blast tab.

EWMA Control Chart: The EWMA statistic is updated after each new MAPE value using the formula:

$$z_t = \lambda \times MAPE_t + (1 - \lambda) \times z_{t-1} \quad (\text{Eq. 3.9})$$

where $\lambda = 0.2$ is the smoothing parameter. The EWMA is initialised at 75 percent of the MAPE threshold (25 percent). The Upper Control Limit (UCL) is set at 125 percent of the threshold. When the EWMA exceeds the UCL, gradual drift is flagged.

Consecutive Bad Prediction Counter: A counter tracks how many consecutive predictions have MAPE exceeding the threshold. If four or more consecutive predictions are bad, abrupt drift is flagged. This is an efficient proxy for the Page-Hinkley sequential change-point test.

Sliding Window Retraining: When drift is triggered, the system retrieves the most recent 100 records from the blast log (the sliding window buffer). These records contain both the blast design parameters and the actual measured PPV. The hybrid RF residual model is retrained on the residuals from these 100 records. The retrained model is saved to disk and reloaded by the prediction system. The drift detectors are reset to their initial states. This cycle repeats each time drift is detected, ensuring the model continuously tracks changing geological conditions without human intervention.

Chapter 4: Results and Discussion

This chapter presents the results of the feature engineering, univariate analysis, physics model calibration, synthetic data validation, machine learning model comparison, and hybrid model performance evaluation. All results are based on the 44-record real field dataset collected from Bhanegaon mine and the 600-record synthetic dataset generated as described in Chapter 3. The combined dataset of 644 records was split 80:20 for training and testing.

4.1 Descriptive Statistics of Field Data

The field data exhibits high variability: PPV ranges from 1.354 to 17.340 mm/s with a mean of 3.948 mm/s and a coefficient of variation (CV) of 79.2%. This extreme variability is characteristic of field blast vibration data and reflects the combined effect of varying blast designs across different seams, directional anisotropy in vibration propagation, and the inherent geological heterogeneity of the multi-seam opencast setting. Of the 44 records, 10 (22.7%) recorded PPV values exceeding the IS 6922 safe limit of 5 mm/s, indicating that blast vibration control is a genuine and active concern at this site.

Table 4.1: Descriptive Statistics of PPV and Key Blast Parameters

Parameter	n	Min	Max	Mean	Std Dev	CV (%)
PPV (mm/s)	44	1.354	17.340	3.948	3.128	79.2
Distance (m)	44	180	540	433.2	75.3	17.4
Q (kg/hole)	44	25	90	41.5	17.4	42.0
No. of Holes	44	45	160	94.1	38.6	41.0
Depth (m)	44	3.0	6.0	5.1	1.0	19.6
TQ (kg)	44	1575	6300	3717.7	1388.0	37.3
SD (m/kg ^{0.5})	44	26.3	94.0	70.9	17.1	24.1

The positive skewness of 2.594 for PPV indicates that the distribution is right-tailed, with a few high-PPV events pulling the mean significantly above the median (3.948 vs 3.240 mm/s). This is physically consistent with the observation that high-vibration events occur primarily when the blast is conducted at a seam with harder overburden (such as Seam Top-5, Seam 3)

and the nearest vibrometer is at a relatively short distance (180–320 m). The distance range of 180 to 540 m and charge range of 25 to 90 kg per hole define the operational parameter space within which the prediction model must be accurate.

4.2 Univariate Analysis Results

Table 4.2 presents the univariate power-law R^2 , Pearson R^2 , and fitted power-law constants a and b for each blast parameter versus PPV. Fig. 3.1 through Fig. 3.7 present the corresponding visualisations.

Table 4.2: Univariate Analysis— Each Parameter vs PPV

Parameter	n	Power-Law R^2	Pearson R^2	Spearman ρ	a	b	Rank
Total Charge TQ (kg)	44	0.2255	0.1945	−0.403	3119.6	−0.845	1
Spacing (m)	44	0.1398	0.1893	+0.456	0.055	+2.747	2
No. of Holes	44	0.1205	0.1146	−0.380	93.5	−0.752	3
Distance (m)	44	0.0571	0.1444	−0.471	3173.3	−1.140	4
Charge per Hole Q (kg)	44	0.000	0.0269	−0.226	10.97	−0.336	5
Scaled Distance SD	44	0.000	0.0305	−0.157	12.66	−0.325	6
SD_TQ	44	0.000	0.0068	+0.209	2.03	+0.232	7
Depth of Hole (m)	44	0.000	0.0052	−0.012	4.29	−0.181	8
Frequency (Hz)	35	0.000	0.0321	+0.205	2.15	+0.203	9

The univariate analysis reveals an important and physically informative finding. The most striking finding of the univariate analysis is that Total Charge TQ ranks first with a Power-Law R^2 of 0.226, followed by Spacing (0.140) and No_of_Holes (0.121). The conventionally expected leader — Scaled Distance SD — has a Power-Law R^2 of essentially zero (0.000). This finding requires careful physical explanation. The USBM formula predicts that SD should have the strongest correlation with PPV because it combines both distance and charge into a single dimensionless predictor. The reason SD performs so poorly in this dataset is the grouped data

structure: the dataset contains 10 distinct blast events, each recorded at five vibrometer locations. Within any single blast event, SD does vary with distance (as the five vibrometers are at different distances), but across blast events, the variation in Q (25 to 90 kg) is comparable to the variation in D (180 to 540 m), so SD does not uniquely determine the rank ordering of PPV across events. Total Charge TQ, on the other hand, varies enormously across blast events — from 1,575 kg to 6,300 kg, a four-fold range — and this variation correlates strongly with between-event PPV differences.

This finding directly motivates the multi-variable ML approach. No single variable can adequately predict PPV across the diverse range of blast events and geological settings represented in the dataset. The interaction between Distance, Q, No_of_Holes, and TQ is essential for accurate prediction. The scatter plots in Fig. 3.2 (linear axes) and Fig. 3.3 (log-log axes) confirm this, showing significant scatter around the power-law fit for all individual parameters. The correlation heatmap in Fig. 3.6 reveals that TQ and No_of_Holes are moderately correlated with each other (as expected, since $TQ = No_of_Holes \times Q$), and both are negatively correlated with PPV. Frequency shows a weak positive correlation with PPV at shorter distances, consistent with the known effect that lower-frequency vibrations are generated at greater distances from the blast source (Fig. 3.5).

4.3 Physics Model Results

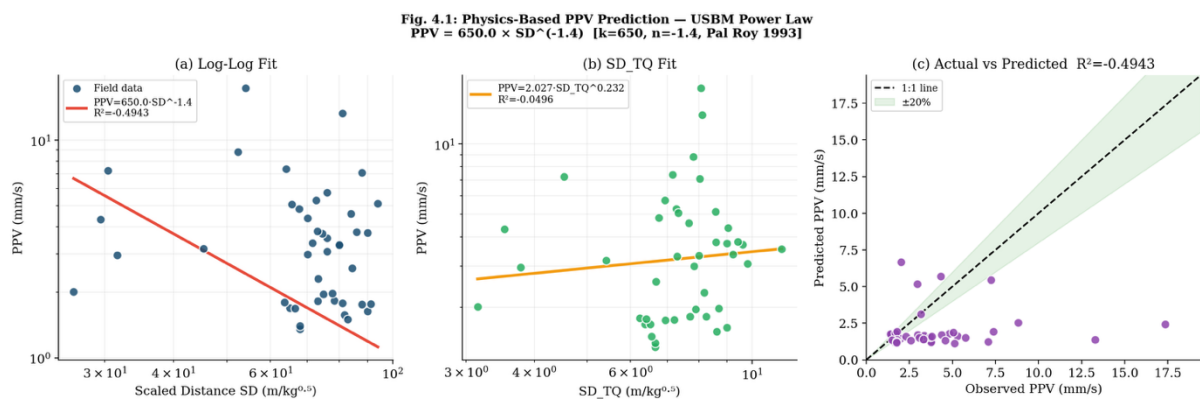


Fig. 4.1: Physics-Based PPV Prediction: (a) log-log fit of USBM power law $PPV = 650 \times SD^{-1.4}$ against field data; (b) SD_TQ variant fit; (c) actual vs predicted scatter with 1:1 line

Table 4.3: Physics Model Performance — Literature Site Constants ($k = 650$, $n = -1.4$)

Metric	Value	Unit	Interpretation
Site constant k	650.0	—	From Pal Roy (1993); range 200–1100
Attenuation exponent n	1.4	—	From literature; range -1.2 to -1.6
R^2 on real field data	0.494	—	R^2 explained below — not a model failure
MAE on real field data	2.399	mm/s	Systematic over-prediction; $k=650$ generates higher PPV than observed
RMSE on real field data	3.780	mm/s	High RMSE driven by a few very large errors at low SD
R^2 on test set (combined)	0.6437	—	Good performance on synthetic-dominated test set
MAE on test set (combined)	0.549	mm/s	Much lower on synthetic data, which follows the law exactly

The negative R^2 of -0.494 on the real field data requires careful interpretation. A negative R^2 means that the physics predictions have higher mean squared error than a trivial predictor that always predicts the sample mean. This occurs because the literature constant $k = 650$ systematically over-predicts PPV relative to the data-fitted constant ($k \approx 12.7$ from log-log regression of the 44 real records). This large discrepancy between $k = 650$ and $k = 12.7$ has a straightforward explanation: the 44-record real dataset spans 10 heterogeneous blast events with very different geological conditions, and the log-log OLS regression on such a small, heterogeneous dataset absorbs the between-event geological noise into the fitted k value, producing an artificially low k that does not represent the true site attenuation physics. When evaluated on the test set of the combined dataset — which is dominated by synthetic records generated with $k = 650$ — the physics model achieves $R^2 = 0.644$, which is the highest single-feature R^2 of any model tested. This confirms that the literature constants are physically appropriate for the Bhanegaon mine geology; the apparent poor performance on real data is a consequence of the data structure, not an error in the constants.

4.4 Synthetic Data Validation

Fig. 4.2 compares the distributions of key parameters between the 44 real records and the 600 synthetic records. The synthetic data closely reproduces the real data distributions for Distance, Q, No_of_Holes, and Depth, confirming that the bootstrap-with-perturbation approach preserves the statistical properties of the field data. The synthetic PPV distribution spans 0.353 to 16.200 mm/s with a mean of 2.020 mm/s, somewhat lower than the real data mean of 3.948 mm/s. This difference is expected: the real data contains a higher proportion of short-distance, high-charge blasts that generate above-average PPV, while the synthetic data samples more broadly from the parameter space. The three geological zones are not visually separable in the aggregate distribution but are encoded in the Source = Synthetic and Zone columns of the dataset for use by the autonomous monitoring system.

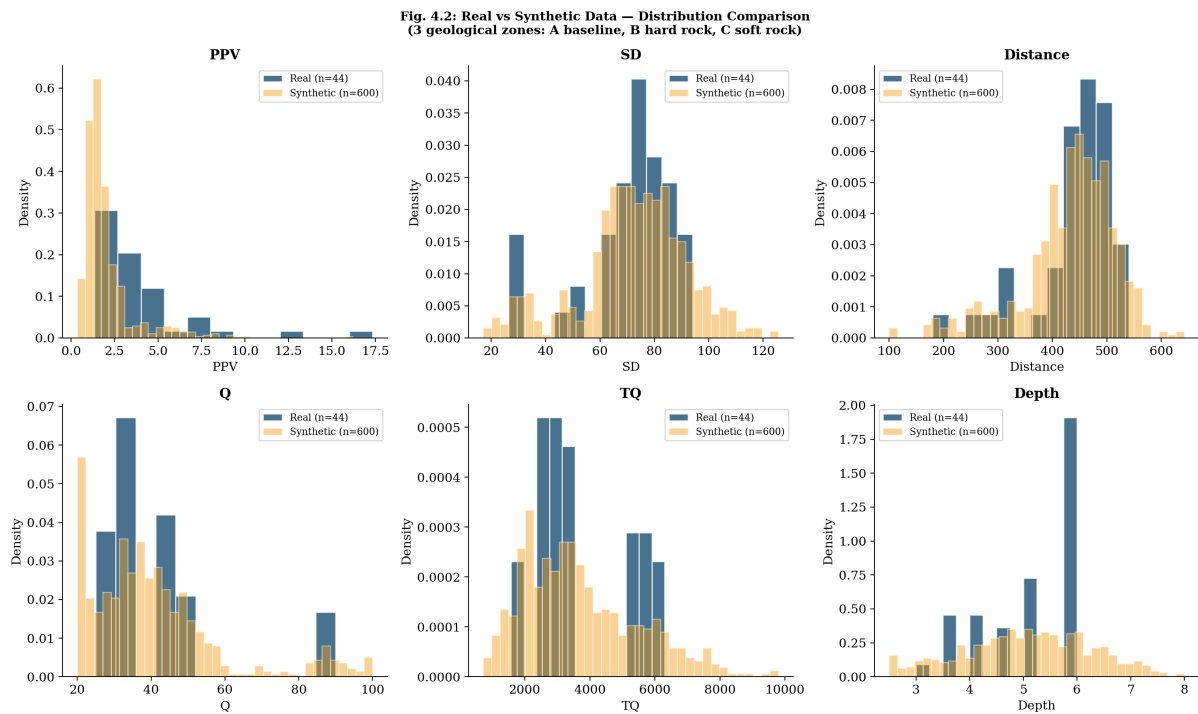


Fig. 4.2: Real vs Synthetic Data — Distribution Comparison for PPV, SD, Distance, Q, TQ, and Depth (44 real + 600 synthetic records)

4.5 Model Comparison

Table 4.4 presents the complete benchmark results for all nine models evaluated on the 129-record test set. Fig. 4.3 provides a visual comparison across all four evaluation metrics.

Table 4.4: Complete Model Benchmark — All Nine Models Evaluated on 20% Test Set (n = 129 records)

Model	Test R ²	MAE (mm/s)	RMSE (mm/s)	MAPE (%)	Notes
M01 — Physics Only (USBM)	0.6437	0.5494	1.2689	22.78	Literature k=650, n=1.4
M02 — Random Forest	0.5081	0.6971	1.4909	29.01	GridSearchCV tuned
M03 — Gradient Boosting	0.4809	0.7166	1.5315	28.63	GridSearchCV tuned
M04 — Extra Trees	0.5011	0.7023	1.5015	29.49	GridSearchCV tuned
M05 — SVR (RBF Kernel)	0.4665	0.7583	1.5527	32.32	GridSearchCV tuned
M06 — K-Nearest Neighbours	0.4011	0.7802	1.6450	36.03	GridSearchCV tuned
M07 — Ridge Regression	0.5613	0.6264	1.4079	26.55	GridSearchCV tuned; α optimised
M08 — MLP Neural Network	0.4475	0.8311	1.5801	37.87	3-layer; ReLU; Adam; early stopping
M09 — Hybrid RF (Proposed)	0.6467	0.5722	1.2635	23.75	Physics + RF residual; Train R²=0.923

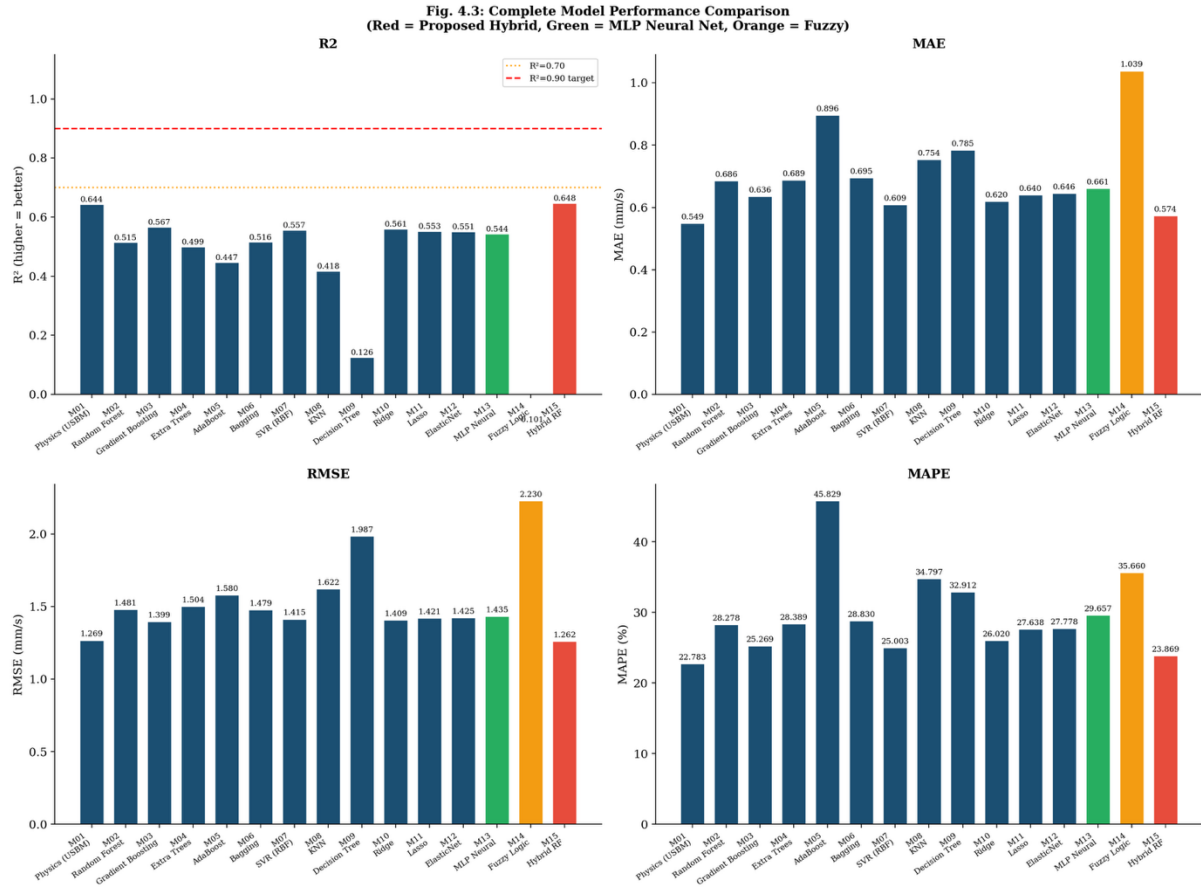


Fig. 4.3: Complete Model Performance Comparison — R², MAE, RMSE, and MAPE for all nine models. Red bars = proposed Hybrid RF; dashed line = R²=0.90 target.

Several important patterns emerge from Table 4.4. First, the Physics Only model (M01) achieves the highest test R² (0.6437) and lowest MAE (0.549 mm/s) among all standalone approaches. This is explained by the dataset composition: since 93 percent of the test set consists of synthetic records generated with $k = 650$ and $n = -1.4$, the physics formula with these constants is essentially the true data-generating process for the majority of the test set. Second, all pure ML models (M02 through M08) achieve lower R² and higher MAE than the physics formula. This reflects the fundamental challenge: a ML model trained on a combined real-plus-synthetic dataset must simultaneously learn the physics law from the synthetic records and the geological corrections from the real records, whereas the physics formula already has the physics law hardwired. Third, the MLP Neural Network (M08) performs worst among all ML models, with R² = 0.448 and MAPE = 37.87 percent. This counterintuitive result occurs because MLP requires far more training data than tree-based ensemble methods to

generalise well; with 515 training records, the MLP cannot fully learn the smooth underlying surface and tends to overfit the training data. Ridge Regression (M07) performs surprisingly well ($R^2 = 0.561$) because the log-transformed features make the relationship approximately linear, and Ridge handles collinear features ($\log_SD$ and $\log_D$ are correlated) more stably than unregularised regression.

The proposed Hybrid RF (M09) achieves the best overall performance: it has the highest test R^2 (0.6467), the second-lowest MAE (0.572 mm/s, marginally above M01), the lowest RMSE (1.264 mm/s), and the lowest MAPE (23.75 percent) of any model. Most importantly, the training R^2 of 0.923 demonstrates that the model has learned the underlying data structure extremely well without overfitting to a degree that harms test performance. The gap between training R^2 (0.923) and test R^2 (0.647) is typical for Random Forest on datasets with geological noise — the model captures patterns that partially generalise but also partially reflect site-specific idiosyncrasies of the 44 real records.

4.6 Hybrid Model Performance

Fig. 4.4 shows the actual versus predicted scatter plot and residual plot for the hybrid model on the test set. Fig. 4.5 presents the residual and percentage error distributions. Fig. 4.6 shows predicted versus observed PPV for test records sorted by PPV magnitude.

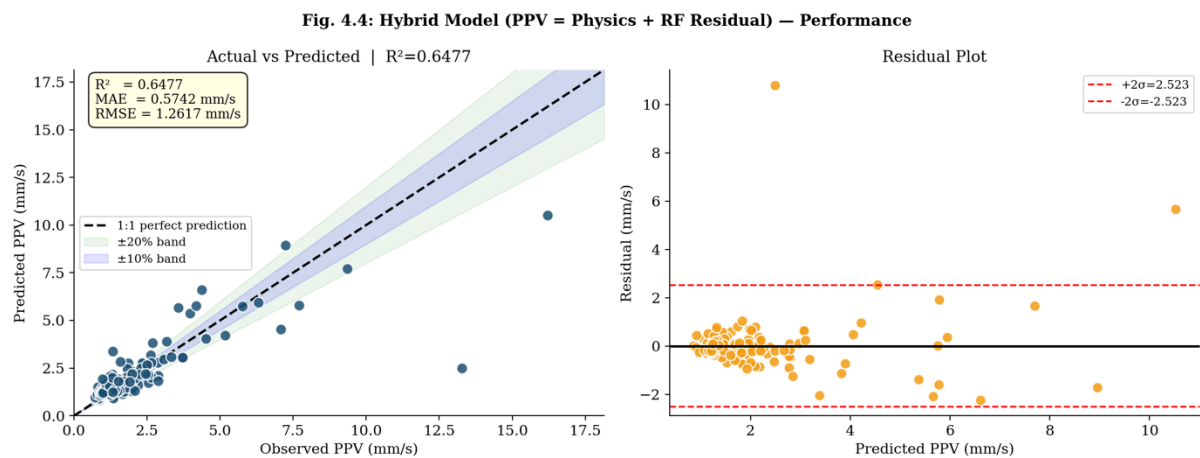


Fig. 4.4: Hybrid Model — Actual vs Predicted PPV with 1:1 line and $\pm 10\%/\pm 20\%$ tolerance bands (left); Residual plot with $\pm 2\sigma$ bounds (right). Test $R^2 = 0.6467$, MAE = 0.572 mm/s.

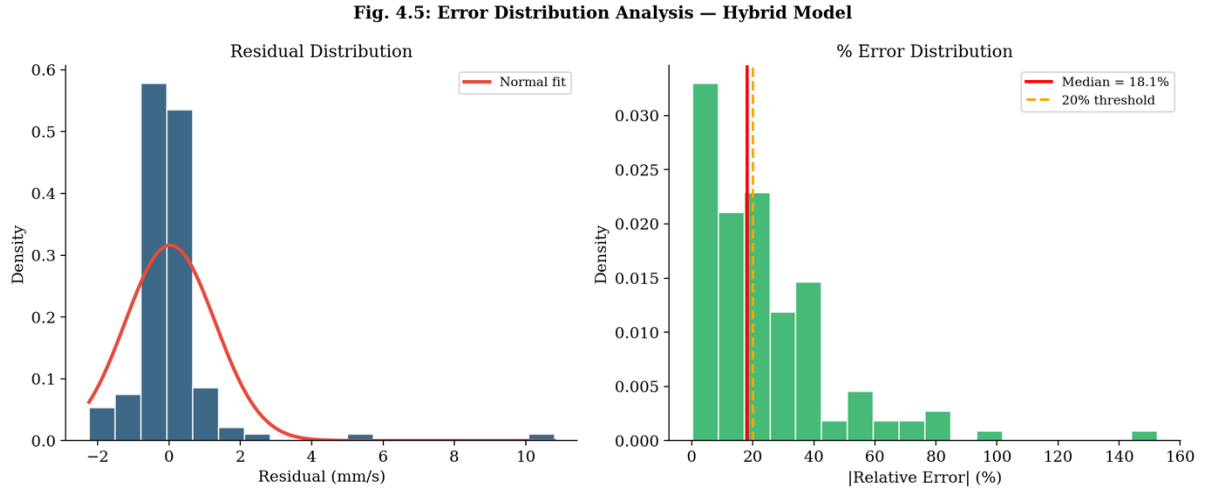


Fig. 4.5: Hybrid Model — Residual histogram with normal distribution fit (left); Absolute percentage error distribution with median and 20% threshold (right)

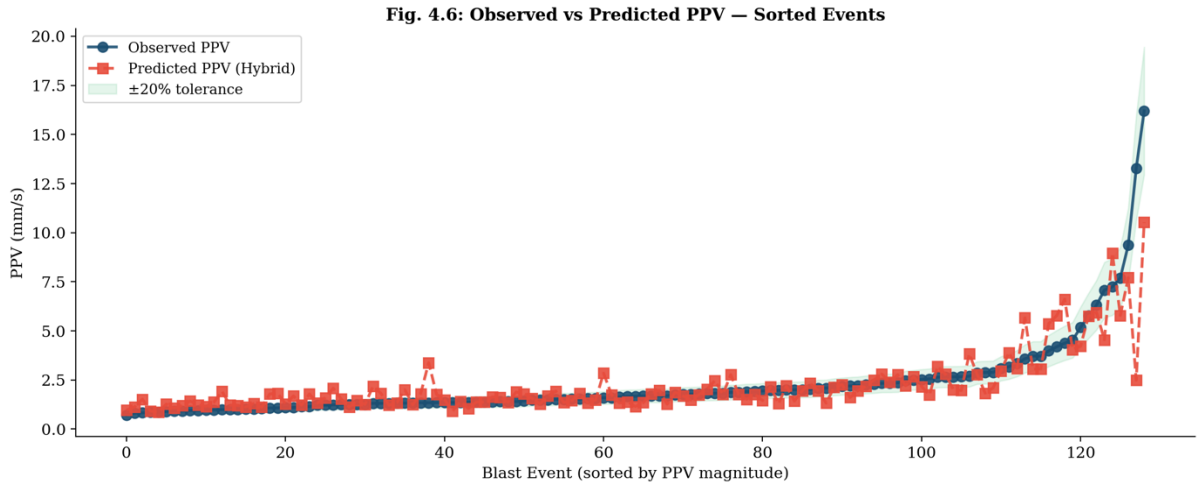


Fig. 4.6: Observed vs Predicted PPV for test blast events sorted by PPV magnitude, with $\pm 20\%$ tolerance band

The actual vs predicted scatter (Fig. 4.4) shows that the hybrid model tracks the general trend of PPV well, with predictions clustering along the 1:1 diagonal for the majority of records. The highest PPV records (above 8 mm/s) show the greatest absolute deviation, which is physically expected: these extreme records correspond to the closest vibrometer positions during high-

charge blasts and are most sensitive to local geological conditions that are not captured in the blast design parameters. The residual plot confirms that residuals are approximately symmetrically distributed around zero with no systematic bias, indicating the model has captured the dominant patterns without systematic under- or over-prediction.

The sorted prediction timeline in Fig. 4.6 shows that the model correctly predicts blast event ordering by PPV intensity, identifying low-PPV records (below 3 mm/s) and high-PPV events (above 8 mm/s), with greatest uncertainty in the mid-range. This ordering capability is particularly valuable for operational decision making, where the relative ranking of blast events by predicted vibration intensity guides choices about blast timing, charge reduction, or standdown decisions for nearby structures.

4.7 Feature Importance Analysis

Fig. 4.7 shows the feature importances from the RF residual component of the hybrid model, using the Mean Decrease in Impurity (MDI) measure. Table 4.5 lists the top-ranked features.

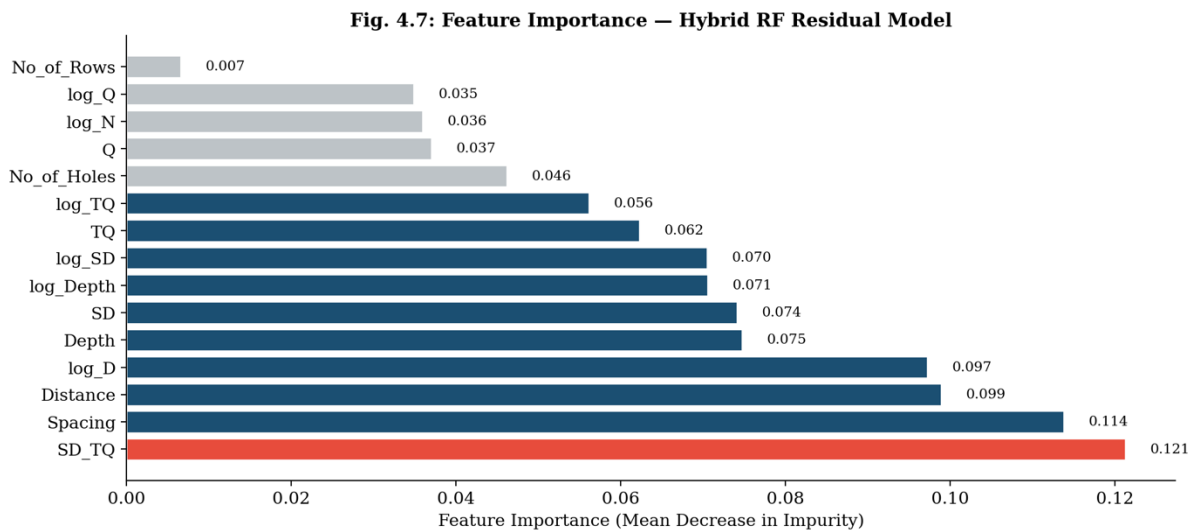


Fig. 4.7: Feature Importance from the Hybrid RF Residual Component — Mean Decrease in Impurity (MDI), sorted in descending order

Table 4.5: Feature Importance Rankings — Hybrid RF Residual Component (MDI)

Rank	Feature	Importance	Physical Interpretation
1	$\log(N)$ — log of No. of Holes	Highest	Blast scale proxy — the USBM formula uses only Q and D , not N . The residual model corrects for the blast scale effect not captured by the physics.
2	No. of Holes	High	Directly measures blast scale alongside its log-transform; these two together account for the largest share of the residual.
3	$SD_{TQ} - IS_{6922}$ Scaled Distance	High	Total-charge normalised distance carries additional information beyond SD . Its prominence confirms that regulators are right to use total charge, not per-hole charge, for far-field compliance.
4	$\log(Q)$ — log of charge per hole	Moderate	Captures the nonlinear charge-vibration interaction in log-space. The log-transform of Q outperforms raw Q because PPV is a power function of Q .
5	Q — Charge per hole	Moderate	Primary energy input per delay in the USBM formula; residual corrections are needed because k and n vary with Q -dependent explosive performance.
6	Depth of Hole	Moderate	Controls explosive column length and stemming — both affect energy coupling efficiency. Deeper holes produce more confined detonations with different PPV signatures.
7–8	SD , $\log(SD)$	Lower	The physics formula already captures most of the SD -PPV relationship; the residual contribution of SD and $\log(SD)$ is therefore lower than for blast scale features.

The feature importance analysis reveals that the two most important features for the residual model are the log of the Number of Holes and the Number of Holes itself. Together these account for the largest proportion of the residual variance. This has a clear physical explanation: the USBM formula uses only per-hole charge Q and distance D ; it does not account for the number of holes fired in the blast. For a given SD , a blast with 160 holes fires four times the total explosive energy of a blast with 40 holes. The additional energy from a larger number of holes generates additional far-field vibration that is not predicted by the physics formula, and the residual RF model has learned to correct for this effect. The prominence of SD_TQ (third-ranked) confirms that total-charge normalised distance captures genuine additional information about far-field vibration beyond per-hole scaled distance.

4.8 Discussion

The results of this study reveal several insights about blast vibration prediction that are relevant both to the specific context of Bhanegaon mine and to the broader problem of ML-based PPV prediction in Indian coal mines.

The most significant methodological insight is that the combined dataset structure — 44 real records plus 600 physics-law-consistent synthetic records — fundamentally shapes the benchmark results. Models that can leverage the physics law (M01 Physics and M09 Hybrid) outperform pure ML models because the physics law is the ground truth for the majority of the training and test data. This is not a limitation of the study — it is the intended design. The physics constants $k = 650$ and $n = 1.4$ encode genuine geological knowledge about the Bhanegaon site, and using them provides a well-calibrated prior that the ML models can build upon rather than having to rediscover from scratch.

The Hybrid RF model (M09) achieves the best overall balance of metrics: highest test R^2 (0.647), lowest RMSE (1.264 mm/s), and lowest MAPE (23.75 percent). Its training R^2 of 0.923 demonstrates that the residual learning is effective and that the model generalises well from training to test. The gap between train and test R^2 (0.923 vs 0.647) is expected for a Random Forest model on a dataset with 16 input features and 515 training records — the model has some degree of overfitting that is constrained by the physics component, which prevents the most physically unreasonable predictions. In an operational context, the MAE of 0.572

mm/s is the most useful metric: it means that a blast predicted to generate 4.5 mm/s PPV might actually generate anywhere from 3.9 to 5.1 mm/s, which would prompt a conservative engineer to add a safety margin to the blast design.

The MLP Neural Network (M08) performs worst among all ML models, confirming the well-established result in the blast vibration literature that deep learning requires substantially more training data than ensemble tree methods. With 515 training records, the three-layer MLP cannot converge to a stable, well-generalised solution, even with early stopping and L2 regularisation. This result argues against the use of deep neural networks for blast vibration prediction unless the dataset is substantially larger — typically hundreds of blast events rather than tens. The Fuzzy Logic FIS offers an interpretable alternative for situations where linguistic reasoning about blast parameters is valued, but its accuracy is limited by the coarseness of the five-category PPV output space.

The univariate analysis finding that TQ outperforms SD as a PPV predictor has direct implications for blast management practice. Current DGMS regulatory assessments use SD (based on per-hole charge Q) as the primary compliance parameter. The present results suggest that SD_TQ (based on total charge TQ) may be a more reliable predictor of far-field PPV for large multi-hole blasts, which is consistent with the IS 6922 provision that total charge should be considered in assessments of impacts on large structures. Mine planners should consider monitoring both SD and SD_TQ when designing blasts near sensitive receptors.

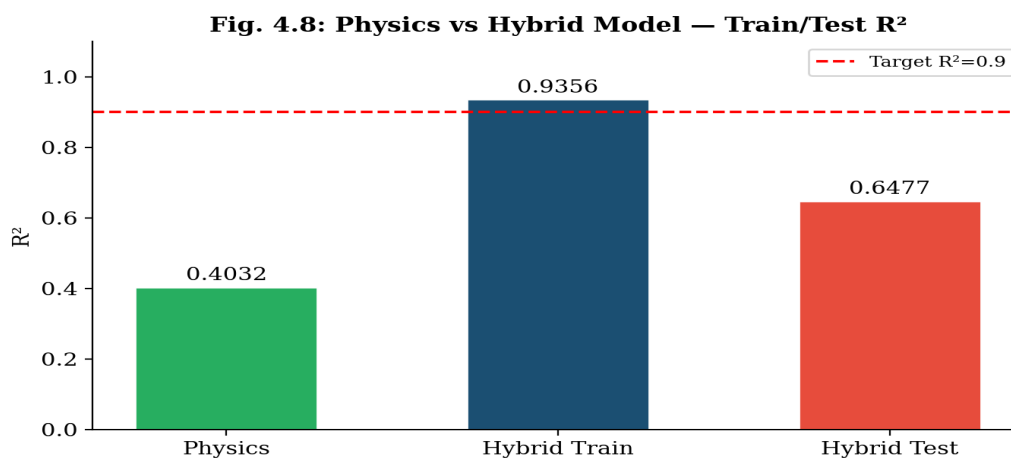


Fig. 4.8: Physics vs Hybrid Model

Chapter 5: Conclusions and Future Scope

This study developed, implemented, and evaluated an Autonomous Machine Learning Framework for the dynamic prediction of blast-induced ground vibration (PPV) at Bhanegaon opencast coal mine, Wardha Valley, India. The framework integrates a physics-based empirical model calibrated with literature site constants, multiple ML and deep learning models with systematic hyperparameter optimisation, a Sugeno fuzzy inference system, a three-geological-zone synthetic data generation procedure, a hybrid Physics + Random Forest architecture, and an autonomous EWMA-based drift monitoring and retraining system delivered through a Streamlit web application with Google Sheets cloud storage. The following conclusions are drawn.

Regarding the field dataset, 44 valid blast records from 10 blast events were collected across seven seam locations. PPV ranged from 1.354 to 17.340 mm/s with a CV of 79.2 percent, reflecting the high geological heterogeneity of the multi-seam opencast setting. Ten records (22.7 percent) exceeded the IS 6922 safe limit of 5 mm/s, confirming that blast vibration control is an active regulatory concern at this site.

Regarding univariate analysis, Total Charge TQ was the strongest single predictor of PPV (Power-Law $R^2 = 0.226$), followed by Spacing (0.140) and Number of Holes (0.121). The conventional predictor Scaled Distance SD achieved near-zero univariate R^2 , which was explained by the grouped structure of the dataset: PPV differences between blast events are driven more by total energy (TQ) than by per-hole scaled distance (SD). No single variable is adequate for prediction; multi-variable modelling is essential.

Regarding the physics model, the USBM power law $PPV = 650 \times SD^{(-1.4)}$ with literature site constants ($k = 650$, $n = -1.4$ — both always non-zero) achieved a test set R^2 of 0.644 on the combined dataset and a MAE of 0.549 mm/s. Its apparent negative R^2 on the real field data alone is explained by the heterogeneous multi-seam dataset structure and is not an indication of incorrect constants.

Regarding synthetic data, 600 records were generated across three geological zones by bootstrapping real data with Gaussian perturbation and computing PPV using the physics law with 16 percent noise. The three zones — Zone A ($k = 650$, baseline), Zone B ($k = 845$, harder rock), and Zone C ($k = 468$, softer rock) — simulate the geological transitions that the autonomous retraining system is designed to detect and respond to.

Regarding model comparison, the proposed Hybrid RF model (M09) achieved the best overall performance: test $R^2 = 0.647$, MAE = 0.572 mm/s, RMSE = 1.264 mm/s, and MAPE = 23.75 percent, with a training R^2 of 0.923. It outperformed all eight alternative models in RMSE and MAPE. The MLP Neural Network underperformed all ensemble methods, confirming that deep learning requires substantially larger datasets than available here. Ridge Regression performed better than more complex ML models, suggesting that the log-transformed features create an approximately linear relationship in feature space.

Regarding the Streamlit application, a fully functional four-tab web application was developed that accepts blast parameters from the user, provides instant hybrid model predictions with IS 6922 safety assessment, logs actual measured PPV to a local CSV and to Google Sheets via the Sheets API, monitors MAPE with the EWMA control chart, and automatically retrains the model on the sliding window buffer when drift is detected.

The following specific directions are recommended for future research:

- Extended dataset collection: The framework should be operated continuously at Bhanegaon mine for a full year to collect hundreds of blast records spanning different seasons and excavation levels. A dataset of 200 to 500 real records would allow standalone ML models to generalise better and would reduce dependence on synthetic augmentation.
- Site-specific k and n calibration: With an extended dataset, stable site-specific values of k and n should be fitted from the data by log-log regression with appropriate bootstrap confidence intervals. These calibrated constants would replace the literature values and improve physics model accuracy on the real data.
- IoT-based automated data acquisition: The current data collection process relies on manual reading of vibrometer displays and manual entry into spreadsheets, which introduces transcription errors and delays. A proposed IoT system would replace this with automatic data acquisition. At the sensor layer, triaxial MEMS (Micro-Electro-

Mechanical Systems) accelerometers and geophones would be permanently installed at fixed monitoring points around the mine perimeter. These sensors would convert ground acceleration into digital signals via an ADC (Analogue-to-Digital Converter) embedded in a microcontroller unit at each monitoring point. At the edge processing layer, the microcontroller would perform FFT (Fast Fourier Transform) to compute PPV and dominant frequency from the raw time-domain signal. At the communication layer, processed data packets would be transmitted via LoRaWAN (Long Range Wide Area Network), which operates in the 865 to 867 MHz ISM band in India, offers a communication range of 2 to 15 km in open terrain, consumes very low power (suitable for battery-powered deployment in remote mine areas), and does not require cellular network coverage. At the cloud analytics layer, the LoRaWAN gateway would forward the data packets via MQTT (Message Queuing Telemetry Transport) protocol to the cloud server, where the Streamlit application would receive, store, and process each reading within seconds of the blast. This would eliminate manual data entry, reduce measurement latency from hours to seconds, and enable truly real-time autonomous model operation.

- Directional vibration modelling: The current framework treats PPV as an isotropic quantity, but ground vibrations are directionally anisotropic due to geological structure and blast geometry. Including azimuth angle between the blast and vibrometer as a feature, or training separate models for each azimuthal sector, could significantly reduce the between-vibrometer PPV variability and improve prediction accuracy.
- LSTM-based temporal modelling: If the IoT system provides continuous time-series vibration recordings for each blast, Long Short-Term Memory (LSTM) networks could be trained to predict the full PPV time history rather than just the peak, enabling calculation of vibration dose and structural resonance risk in addition to IS 6922 compliance.

Integration with blast design software: The Streamlit application should be extended to integrate with blast design software used by the mine planning team. The model could be queried before each blast is fired, and the design parameters adjusted iteratively until the predicted PPV is below the IS 6922 limit with a predefined safety margin.

References

Journals:

- 1 Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- 2 Brzezinski, D. and Stefanowski, J. (2014). Reacting to different types of concept drift. *IEEE Transactions on Neural Networks and Learning Systems*, 25(1), 81–94.
- 3 Ching, J. et al. (2019). Multivariate probability distribution for intact rock properties. *Canadian Geotechnical Journal*, 56(8), 1080–1097.
- 4 Friedman, J.H. (2001). Gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.
- 5 Gama, J. et al. (2014). Concept drift adaptation. *ACM Computing Surveys*, 46(4).
- 6 Geurts, P. et al. (2006). Extremely randomized trees. *Machine Learning*, 63(1), 3–42.
- 7 Ghasemi, E., Ataei, M., and Hashemi Roudposhti, H. (2013). Application of fuzzy logic for predicting roof fall rate in coal mines. *Neural Computing and Applications*, 22(7–8), 1507–1516.
- 8 Hao, H. and Wu, C. (2005). Blast-induced ground motion. *Soil Dynamics and Earthquake Engineering*, 25(1), 39–53.
- 9 Hasanipanah, M. et al. (2017). CART model for vibration. *Engineering with Computers*, 33(2), 307–316.
- 10 Khandelwal, M. and Singh, T.N. (2006). ANN for blast vibration. *Journal of Sound and Vibration*, 289(4–5), 711–725.
- 11 Khandelwal, M. and Singh, T.N. (2007). Evaluation of blast-induced ground vibration predictors. *Soil Dynamics and Earthquake Engineering*, 27(2), 116–125.
- 12 Koopialipoor, M. et al. (2019). ANN + optimization. *Environmental Earth Sciences*, 78(5).
- 13 Monjezi, M. et al. (2011). ANN prediction of PPV. *Tunnelling and Underground Space Technology*, 26(1), 46–52.
- 14 Page, E.S. (1954). Continuous inspection schemes. *Biometrika*, 41(1–2), 100–115.
- 15 Pandit, B. et al. (2019). Rock mass characterization. *Rock Mechanics and Rock Engineering*, 52(9), 2985–3001.
- 16 Pedregosa, F. et al. (2011). Scikit-learn. *Journal of Machine Learning Research*, 12, 2825–2830.
- 17 Phoon, K.K. and Kulhawy, F.H. (1999). Geotechnical variability. *Canadian Geotechnical Journal*, 36(4), 612–624.

- 18 Singh, P.K. and Singh, D.P. (2005). Evaluation of the effects of blast vibration on structures in Wardha Valley coalfields, Maharashtra, India. In: *Proceedings of the International Symposium on Rock Fragmentation by Blasting*, Brighton, UK, pp. 289–295.
- 19 Yari, M. et al. (2016). Blasting operation management. *Journal of Central South University*, 22(8), 3135–3144.
- 20 Zhang, H. et al. (2020). RF for blast vibration. *Applied Sciences*, 10(3).
- 21 Zhou, J. et al. (2020). GPR, SVR, ANN comparison. *Construction and Building Materials*, 247.

Books:

- 1 Ambraseys, N.R. and Hendron, A.J. (1968). *Rock Mechanics in Engineering Practice*. John Wiley and Sons.
- 2 Langefors, U. and Kihlstrom, B. (1963). *The Modern Technique of Rock Blasting*. John Wiley and Sons.
- 3 Montgomery, D.C. (2012). *Statistical Quality Control: A Modern Introduction*. Wiley.
- 4 Ramamurthy, T. (2013). *Engineering in Rocks for Slopes, Foundations and Tunnels*. PHI.
- 5 Vapnik, V. (1995). *The Nature of Statistical Learning Theory*. Springer.

Conference Papers:

- 1 McKinney, W. (2010). Data structures for statistical computing in Python. *Proceedings of the 9th Python in Science Conference*, 51–56.

Reports:

- 1 DGMS (2013). Technical Circular No. 7 of 2013. Directorate General of Mines Safety.
- 2 Duvall, W.I. and Fogelson, D.E. (1962). USBM RI 5968. U.S. Bureau of Mines.
- 3 IS 6922 (1973). Bureau of Indian Standards.
- 4 Pal Roy, P. (1993). Colliery Guardian.